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INTERNAL NOTE

MATHEMATICAL OBSERVATION MODELS

M. Jones, T. Hubinek, M. Barbier

Abstract

Description of the new mathematical observation models for LGC2.

Mots-clés : LGC2, modélisation des observations

Key words : LGC2, mathematical observation models

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1. GENERAL MATHEMATIC MODEL

Survey instrument or target:

- position defined in the CCS, or a Local Object Reference (LOR) system
- orientation defined with respect to a Modified Local Astronomical (MLA) system, or a LOR system

1.1 Generic observation equation

The mathematical model for an observation can be represented by the following function,

$$\boldsymbol{F}\left(\underline{\boldsymbol{x}}, \underline{\boldsymbol{l}}\right) = 0$$

where,

$\underline{\boldsymbol{x}} = (x_1, \dots, x_u)$ — vector of unknown parameters
$\underline{\boldsymbol{l}} = (l_1, \dots, l_n)$ — vector of observations

Alternatively this could be broken down further to give,

$$\boldsymbol{F}_L\left(\underline{\boldsymbol{x}_{\boldsymbol{L}}}, \underline{\boldsymbol{l}}\right) - \boldsymbol{F}_R\left(\underline{\Delta\boldsymbol{X}}, \underline{\boldsymbol{x}_{\boldsymbol{R}}}\right) = 0$$

where,

$\boldsymbol{x}_K$ — Observation Unknowns
$\boldsymbol{x}_R$ — Observation Reference Frame Unknowns

and □X is a function of several parameters,

$$\underline{\Delta\boldsymbol{X}} = \boldsymbol{F}_{\boldsymbol{P}}\left(\boldsymbol{X}_T, \underline{\boldsymbol{H}}_T, \boldsymbol{X}_S, \underline{\boldsymbol{H}}_I\right)$$

where

$\boldsymbol{X}_T$ — target position in Observation Reference Frame
$\boldsymbol{X}_S$ — station position in Observation Reference Frame
$\underline{\boldsymbol{H}}_T$— target height
$\underline{\boldsymbol{H}}_I$— instrument height

Now the point coordinates must be transformed into the Observation Reference Frame. Let then be described by the following transformation functions,

$$\boldsymbol{X}_T = \boldsymbol{M}_T\left(\underline{\boldsymbol{x}}_T\right)$$
$$\boldsymbol{X}_S = \boldsymbol{M}_S\left(\underline{\boldsymbol{x}}_S\right)$$

Substituting for X_T and X_S gives,

$$\underline{\Delta\boldsymbol{X}} = \boldsymbol{F}_{\boldsymbol{P}}\left(\boldsymbol{M}_T\left(\underline{\boldsymbol{x}}_T\right), \underline{\boldsymbol{H}}_T, \boldsymbol{M}_S\left(\underline{\boldsymbol{x}}_S\right), \underline{\boldsymbol{H}}_I\right)$$

1.2 Point transformations

1.2.1 Local Reference Frame Tranformations

Let us first consider the transformation between two Cartesian local object reference frame *LOR1* and *LOR2*, where the transformation between any two related frames is a simple Helmert transformation, and let the transformation from *LOR1* to *LOR2* involve a series of transformations through connected reference frames.

1.2.1.1 Cartesian Coordinate Transformation

In Cartesian coordinates the transformation between the reference frames can be defined as follows,

$$\boldsymbol{x}_{LOR1} = \boldsymbol{x}_0 = l_0 \cdot \boldsymbol{R}_0 \cdot \boldsymbol{x}_0 + \boldsymbol{T}_0$$

$$\boldsymbol{x}_1 = l_1 \cdot \boldsymbol{R}_1 \cdot \boldsymbol{x}_0 + \boldsymbol{T}_1$$

and,

$$\boldsymbol{X}_{LOR2} = \boldsymbol{x}_n = l_n \cdot \boldsymbol{R}_n \cdot \boldsymbol{x}_{n-1} + \boldsymbol{T}_n$$

where,

$$\boldsymbol{x}_i = point\ coordinate\ vector$$
$$l_i = transformation\ scale\ factor$$

$$\boldsymbol{R}_i(\omega, \varphi, \kappa) = 3D\ Rotation\ Matrix = \begin{pmatrix} r_{i,11} & r_{i,12} & r_{i,13} \\ r_{i,21} & r_{i,22} & r_{i,23} \\ r_{i,31} & r_{i,32} & r_{i,33} \end{pmatrix}$$

$$\boldsymbol{T}_i = 3D\ translation\ vector = \begin{pmatrix} t_{i,1} \\ t_{i,2} \\ t_{i,3} \end{pmatrix}$$

and,

$$l_0 = 1,\ \boldsymbol{R}_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \boldsymbol{T}_0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

We can therefore write,

$$\boldsymbol{X}_{LOR2} = \boldsymbol{x}_n = \prod_{i=n}^{0} l_i \cdot \boldsymbol{R}_i \cdot \boldsymbol{x}_0 + \sum_{m=1}^{n} \left(\prod_{i=n}^{m} l_i \cdot \boldsymbol{R}_i \right) \cdot \boldsymbol{T}_{m-1} + \boldsymbol{T}_n$$

1.2.1.2 Homogeneous Coordinates Transformation

In homogeneous coordinates the transformation between the reference frames can be defined more simply as follows,

Let the points have homogeneous coordinates,

$$\boldsymbol{x}_i^{\boldsymbol{h}} = \begin{bmatrix} x_i^h \\ y_i^h \\ z_i^h \\ w_i \end{bmatrix}$$

And let,

$$\boldsymbol{x}_0^{\boldsymbol{h}} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \\ 1 \end{bmatrix}$$

Then we can write,

$$\boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = \prod_{i=n}^{0} \boldsymbol{M}_i \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

Where,

$$\boldsymbol{M}_i = \begin{bmatrix} r_{i,11} & r_{i,12} & r_{i,13} & t_{i,1}/l_i \\ r_{i,21} & r_{i,22} & r_{i,23} & t_{i,2}/l_i \\ r_{i,31} & r_{i,32} & r_{i,33} & t_{i,3}/l_i \\ 0 & 0 & 0 & 1/l_i \end{bmatrix}$$

Let,

$$\boldsymbol{M}_{j,m} = \prod_{i=j}^{m} \boldsymbol{M}_i = \begin{bmatrix} M_{jm,11} & M_{jm,12} & M_{jm,13} & M_{jm,14} \\ M_{jm,21} & M_{jm,22} & M_{jm,23} & M_{jm,24} \\ M_{jm,31} & M_{jm,32} & M_{jm,33} & M_{jm,34} \\ 0 & 0 & 0 & M_{jm,44} \end{bmatrix}$$

Then,

$$\boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = \boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

Above, the general transformation model was described. However, from implementation point of view, it is necessary to define direction in the tree, which includes local frames, and to distinguish between paths up and down.
We define *step up* to be represented by$\boldsymbol{M_i}$ and *step down* $\boldsymbol{M_i^{-1}}$ by (which is inverse of the step up) and introduce following labelling for determining path in the tree:

${}_{\mathrm{P}}\boldsymbol{M}_0$— Identity transformation in the Point object reference.
${}_{\mathrm{P}}\boldsymbol{M}_i$ — *i*-th *step up* from the Point object reference (path up is unambiguous).
${}_{\mathrm{I}}\boldsymbol{M}_0{}^{-1}$— Identity transformation in the Instrument object reference.
${}_{\mathrm{I}}\boldsymbol{M}_i^{-1}$ — Inverse of *i*-th *step up* from the Instrument object reference (i.e. its *step down*).
${}_{\mathrm{I}}\boldsymbol{M}_i$—*i*-th *step up* from the Instrument object reference (path up is unambiguous).

Considering this, for a path which gives us from the Point local reference to the Instrument local reference, we get (see illustration on Figure below),

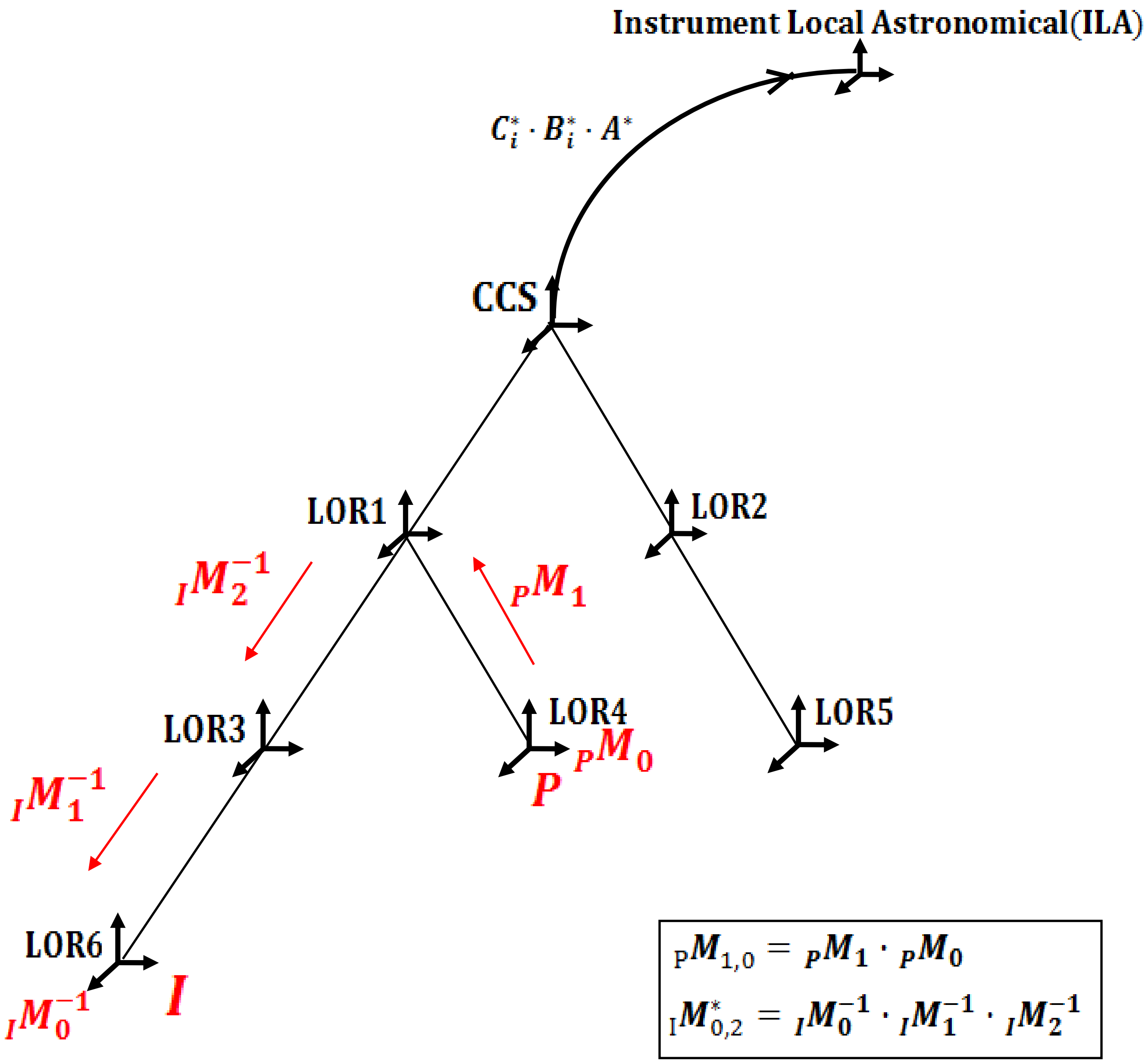
- *n*-steps up,

$${}_{\mathrm{P}}\boldsymbol{M}_{n,0} = \prod_{i=n}^{0} {}_{\mathrm{P}}\boldsymbol{M}_i$$

- *m*-steps down,

$${}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} = \prod_{i=0}^{m} {}_{\mathrm{I}}\boldsymbol{M}_i{}^{-1}$$

Note that transformation to CCS requires only steps up.



Then,

$$\boldsymbol{M} = {_\mathrm{I}\boldsymbol{M}^*_{0,m}} \cdot {_\mathrm{P}\boldsymbol{M}_{n,0}} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ 0 & 0 & 0 & M_{44} \end{bmatrix} \tag{1.1}$$

Finally,

$$\boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_{0}^{\boldsymbol{h}} = \boldsymbol{M} \cdot \boldsymbol{x}_{0}^{\boldsymbol{h}}$$

$$\boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = \begin{bmatrix} M_{11}x_0 + M_{12}y_0 + M_{13}z_0 + M_{14} \\ M_{21}x_0 + M_{22}y_0 + M_{23}z_0 + M_{24} \\ M_{31}x_0 + M_{32}y_0 + M_{33}z_0 + M_{34} \\ M_{44} \end{bmatrix} = \begin{bmatrix} \frac{1}{L} \cdot X_{LOR2} \\ \frac{1}{L} \cdot Y_{LOR2} \\ \frac{1}{L} \cdot Z_{LOR2} \\ \frac{1}{L} \end{bmatrix} = \frac{1}{L} \cdot \begin{bmatrix} X_{LOR2} \\ Y_{LOR2} \\ Z_{LOR2} \\ 1 \end{bmatrix}$$

$L = {}_{\mathrm{I}}l_0^{*} \cdot \ldots \cdot {}_{\mathrm{I}}l_m^{*} \cdot {}_{P}l_n \cdot \ldots \cdot {}_{P}l_0$, is the transformation scale factor.

In Cartesian coordinates form (first three coordinates represent point in Cartesian c.), we have:

$$\boldsymbol{X}_{LOR2} = L \cdot \boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = \begin{bmatrix} X_{LOR2} \\ Y_{LOR2} \\ Z_{LOR2} \\ 1 \end{bmatrix}$$

Giving the following partial derivatives,

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial x_0} = L \cdot \begin{bmatrix} M_{11} \\ M_{21} \\ M_{31} \\ 0 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial y_0} = L \cdot \begin{bmatrix} M_{12} \\ M_{22} \\ M_{32} \\ 0 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial z_0} = L \cdot \begin{bmatrix} M_{13} \\ M_{23} \\ M_{33} \\ 0 \end{bmatrix}$$

1.2.1.3 Direct transformation: transformation from the Point reference system (step up)

Following the convention [1], a transformation is written as below:

$$M_i = \begin{pmatrix} \cos\kappa_i \cdot \cos\varphi_i & \sin\kappa_i \cdot \cos\varphi_i & -\sin\varphi_i & t_1/l_i \\ -\sin\kappa_i \cdot \cos\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i & \cos\kappa_i \cdot \cos\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i & \cos\varphi_i \cdot \sin\omega_i & t_2/l_i \\ \sin\kappa_i \cdot \sin\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i & -\cos\kappa_i \cdot \sin\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i & \cos\varphi_i \cdot \cos\omega_i & t_3/l_i \\ 0 & 0 & 0 & 1/l_i \end{pmatrix}$$

For the transformation parameters we note that,

For $\mathbf{1 \le k < n}$,

$$_{\mathrm{P}}\boldsymbol{M}_{n,0} = {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k,k} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0}$$
$$\boldsymbol{X}_{LOR2}^{\boldsymbol{h}} = {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot \left({}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k,k} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0}\right) \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

In Cartesian coordinates form,

$$\begin{aligned}\boldsymbol{X_{LOR2}} &= ({}_{\mathrm{I}}l_0^* \cdot \dots \cdot {}_{\mathrm{I}}l_m^*) \cdot \left({}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*}\right) \cdot ({}_{P}l_n \cdot \dots \cdot {}_{P}l_0) \cdot \left({}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k,k} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0}\right) \cdot \boldsymbol{x}_0^{\boldsymbol{h}} \\ &= ({}_{\mathrm{I}}l_0^* \cdot \dots \cdot {}_{\mathrm{I}}l_m^* \cdot {}_{P}l_n \cdot \dots \cdot {}_{P}l_0) \cdot \left({}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*}\right) \cdot \left({}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k,k} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0}\right) \cdot \boldsymbol{x}_0^{\boldsymbol{h}} \\ &= L \cdot \left({}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*}\right) \cdot \left({}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k,k} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0}\right) \cdot \boldsymbol{x}_0^{\boldsymbol{h}}\end{aligned}$$

If $\mathbf{k = n}$, then

$$_{\mathrm{P}}\boldsymbol{M}_{n,k+1} = identity\ transformation = \boldsymbol{I}$$

Case $\mathbf{k = 0}$ is not considered, since it is an identity transformation.
Giving the following partial derivatives,

<u>*Angle partial derivatives:*</u>

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial {}_{P}\omega_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} \frac{\partial {}_{P}r_{k,11}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,12}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,13}}{\partial {}_{P}\omega_k} & 0 \\ \frac{\partial {}_{P}r_{k,21}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,22}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,23}}{\partial {}_{P}\omega_k} & 0 \\ \frac{\partial {}_{P}r_{k,31}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,32}}{\partial {}_{P}\omega_k} & \frac{\partial {}_{P}r_{k,33}}{\partial {}_{P}\omega_k} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}} =$$
$$= L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot$$
$$\cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ \sin {}_{P}\omega_k \cdot \sin {}_{P}\kappa_k + \cos {}_{P}\omega_k \cdot \cos {}_{P}\kappa_k \cdot \sin {}_{P}\varphi_k & -\sin {}_{P}\omega_k \cdot \cos {}_{P}\kappa_k + \cos {}_{P}\omega_k \cdot \sin {}_{P}\kappa_k \cdot \sin {}_{P}\varphi_k & \cos {}_{P}\omega_k \cdot \cos {}_{P}\varphi_k & 0 \\ \cos {}_{P}\omega_k \cdot \sin {}_{P}\kappa_k - \sin {}_{P}\omega_k \cdot \cos {}_{P}\kappa_k \cdot \sin {}_{P}\varphi_k & -\cos {}_{P}\omega_k \cdot \cos {}_{P}\kappa_k - \sin {}_{P}\omega_k \cdot \sin {}_{P}\kappa_k \cdot \sin {}_{P}\varphi_k & -\sin {}_{P}\omega_k \cdot \cos {}_{P}\varphi_k & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot$$
$$\cdot \; {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial {}_{P}\varphi_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} \frac{\partial {}_{P}r_{k,11}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,12}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,13}}{\partial {}_{P}\varphi_k} & 0 \\ \frac{\partial {}_{P}r_{k,21}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,22}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,23}}{\partial {}_{P}\varphi_k} & 0 \\ \frac{\partial {}_{P}r_{k,31}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,32}}{\partial {}_{P}\varphi_k} & \frac{\partial {}_{P}r_{k,33}}{\partial {}_{P}\varphi_k} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}} =$$
$$= L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,m}^{*} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1}.$$
$$\cdot \begin{bmatrix} -\cos {}_{P}\kappa_k \,.\, \sin {}_{P}\varphi_k & -\sin {}_{P}\kappa_k \,.\, \sin {}_{P}\varphi_k & -\cos {}_{P}\varphi_k & 0 \\ \cos {}_{P}\kappa_k \cdot \cos {}_{P}\varphi_k \,.\, \sin {}_{P}\omega_k & \sin {}_{P}\kappa_k \cdot \cos {}_{P}\varphi_k \,.\, \sin {}_{P}\omega_k & -\sin {}_{P}\omega_k \cdot \sin {}_{P}\varphi_k & 0 \\ \cos {}_{P}\kappa_k \cdot \cos {}_{P}\varphi_k \,.\, \cos {}_{P}\omega_k & \sin {}_{P}\kappa_k \cdot \cos {}_{P}\varphi_k \,.\, \cos {}_{P}\omega_k & -\cos {}_{P}\omega_k \cdot \sin {}_{P}\varphi_k & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$
$$\cdot \; {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_P\kappa_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} \frac{\partial_P r_{k,11}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,12}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,13}}{\partial_P\kappa_k} & 0 \\ \frac{\partial_P r_{k,21}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,22}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,23}}{\partial_P\kappa_k} & 0 \\ \frac{\partial_P r_{k,31}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,32}}{\partial_P\kappa_k} & \frac{\partial_P r_{k,33}}{\partial_P\kappa_k} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}} =$$

$$= L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot$$

$$\cdot \begin{bmatrix} -\sin {}_P\kappa_k \,.\cos {}_P\varphi_k & \cos {}_P\kappa_k \,.\cos {}_P\varphi_k & 0 & 0 \\ -\sin {}_P\kappa_k \cdot \sin {}_P\varphi_k \,.\sin {}_P\omega_k - \cos {}_P\kappa_k \,.\cos {}_P\omega_k & \cos {}_P\kappa_k \cdot \sin {}_P\varphi_k \,.\sin {}_P\omega_k - \sin {}_P\kappa_k \,.\cos {}_P\omega_k & 0 & 0 \\ -\sin {}_P\kappa_k \cdot \sin {}_P\varphi_k \,.\cos {}_P\omega_k + \cos {}_P\kappa_k \,.\sin {}_P\omega_k & \cos {}_P\kappa_k \cdot \sin {}_P\varphi_k \,.\cos {}_P\omega_k + \sin {}_P\kappa_k \,.\sin {}_P\omega_k & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} .$$

$$\cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

<u>*Translation partial derivatives:*</u>

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_P t_{k,1}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} 0 & 0 & 0 & 1/{}_P l_k \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot\cdot\, {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_P t_{k,2}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/{}_P l_k \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot\cdot\, {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_P t_{k,3}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/{}_P l_k \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot\cdot\, {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

<u>*Scale factor partial derivatives:*</u>

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_P l_k} = ({}_{\mathrm{I}}l_0^* \cdot \ldots \cdot {}_{\mathrm{I}}l_m^* \cdot {}_P l_n \cdot \ldots \cdot {}_P l_{k+1} \cdot 1 \cdot {}_P l_{k-1} \cdot \ldots {}_P l_0) \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,k+1} \cdot \begin{bmatrix} 0 & 0 & 0 & \frac{-1}{({}_P l_k)^2} \\ 0 & 0 & 0 & \frac{1}{({}_P l_k)^2} \\ 0 & 0 & 0 & \frac{-1}{({}_P l_k)^2} \\ 0 & 0 & 0 & \frac{-1}{({}_P l_k)^2} \end{bmatrix} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{k-1,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

1.2.1.4 Inverse transformation: transformation to the Instrument reference system (step down)

For $\mathbf{1 \leq k < m}$,

$$
\begin{gathered}
{}_{\mathrm{I}}\boldsymbol{M}^*_{0,m} = {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k,k} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \\
\boldsymbol{X}^{\boldsymbol{h}}_{LOR2} = \left({}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k,k} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m}\right) \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}^{\boldsymbol{h}}_0
\end{gathered}
$$

In Cartesian form,

$$
\begin{aligned}
\boldsymbol{X}_{LOR2} &= ({}_{\mathrm{I}}l^*_0 \cdot \ldots \cdot {}_{\mathrm{I}}l^*_m) \cdot \left({}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k,k} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m}\right) \cdot ({}_{P}l_n \cdot \ldots \cdot {}_{P}l_0) \cdot \left({}_{\mathrm{P}}\boldsymbol{M}_{n,0}\right) \cdot \boldsymbol{x}^{\boldsymbol{h}}_0 = \\
&= ({}_{\mathrm{I}}l^*_0 \cdot \ldots \cdot {}_{\mathrm{I}}l^*_m \cdot {}_{P}l_n \cdot \ldots \cdot {}_{P}l_0) \cdot \left({}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k,k} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m}\right) \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}^{\boldsymbol{h}}_0 \\
&= L \cdot \left({}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k,k} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m}\right) \cdot ({}_{\mathrm{P}}\boldsymbol{M}_{n,0}) \cdot \boldsymbol{x}^{\boldsymbol{h}}_0
\end{aligned}
$$

If $\mathbf{k = m}$, then

$$
{}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} = identity\ transformation = \boldsymbol{I}
$$

Case $\mathbf{k = 0}$ is not considered, since it is an identity transformation.
Note that inverse transformation of,

$$
\boldsymbol{M}_{i,i} = \boldsymbol{M}_i = \begin{bmatrix} r_{i,11} & r_{i,12} & r_{i,13} & t_{i,1}/l_i \\ r_{i,21} & r_{i,22} & r_{i,23} & t_{i,2}/l_i \\ r_{i,31} & r_{i,32} & r_{i,33} & t_{i,3}/l_i \\ 0 & 0 & 0 & 1/l_i \end{bmatrix}
$$

Is,

$$
\boldsymbol{M}^*_{i,i} = \boldsymbol{M}^{-1}_i = \begin{bmatrix} r^*_{i,11} & r^*_{i,12} & r^*_{i,13} & t^*_{i.1} \\ r^*_{i,21} & r^*_{i,22} & r^*_{i,23} & t^*_{i.2} \\ r^*_{i,31} & r^*_{i,32} & r^*_{i,33} & t^*_{i.2} \\ 0 & 0 & 0 & {}^1/_{l^*_i} \end{bmatrix} = \begin{bmatrix} r_{i,11} & r_{i,21} & r_{i,31} & -(r_{i,11} \cdot t_{i,1} + r_{i,21} \cdot t_{i,2} + r_{i,31} \cdot t_{i,3}) \\ r_{i,12} & r_{i,22} & r_{i,32} & -(r_{i,12} \cdot t_{i,1} + r_{i,22} \cdot t_{i,2} + r_{i,32} \cdot t_{i,3}) \\ r_{i,13} & r_{i,23} & r_{i,33} & -(r_{i,13} \cdot t_{i,1} + r_{i,23} \cdot t_{i,2} + r_{i,33} \cdot t_{i,3}) \\ 0 & 0 & 0 & l_i \end{bmatrix}
$$

To simplify the inverse transformation is written with the same parameters as the direct transformation.

Let,

$$
R^{-1} = \begin{pmatrix} \cos\kappa_i \cdot \cos\varphi_i & -\sin\kappa_i \cdot \cos\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i & \sin\kappa_i \cdot \sin\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i \\ \sin\kappa_i \cdot \cos\varphi_i & \cos\kappa_i \cdot \cos\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i & -\cos\kappa_i \cdot \sin\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i \\ -\sin\varphi_i & \cos\varphi_i \cdot \sin\omega_i & \cos\varphi_i \cdot \cos\omega_i \end{pmatrix}
$$

Angle partial derivatives:

$$\frac{\partial_{\mathrm{I}}\boldsymbol{R}_k^{-1}}{\partial_{\mathrm{I}}\omega_k} = \begin{pmatrix} \frac{\partial_{\mathrm{I}} r_{k,11}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,12}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,13}^*}{\partial_{\mathrm{I}}\omega_k} \\ \frac{\partial_{\mathrm{I}} r_{k,21}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,22}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,23}^*}{\partial_{\mathrm{I}}\omega_k} \\ \frac{\partial_{\mathrm{I}} r_{k,31}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,32}^*}{\partial_{\mathrm{I}}\omega_k} & \frac{\partial_{\mathrm{I}} r_{k,33}^*}{\partial_{\mathrm{I}}\omega_k} \end{pmatrix} = \begin{pmatrix} \mathrm{RO}_{11} & \mathrm{RO}_{12} & \mathrm{RO}_{13} \\ \mathrm{RO}_{21} & \mathrm{RO}_{22} & \mathrm{RO}_{23} \\ \mathrm{RO}_{31} & \mathrm{RO}_{32} & \mathrm{RO}_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} 0 & \sin{}_{\mathrm{I}}\omega_k \cdot \sin{}_{\mathrm{I}}\kappa_k + \cos{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k & \cos{}_{\mathrm{I}}\omega_k \cdot \sin{}_{\mathrm{I}}\kappa_k - \sin{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \\ 0 & -\sin{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\kappa_k + \cos{}_{\mathrm{I}}\omega_k \cdot \sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k & -\cos{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\kappa_k - \sin{}_{\mathrm{I}}\omega_k \cdot \sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \\ 0 & \cos{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\varphi_k & -\sin{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\varphi_k \end{pmatrix}$$

$$\frac{\partial_{\mathrm{I}}\boldsymbol{R}_k^{-1}}{\partial_{\mathrm{I}}\varphi_k} = \begin{pmatrix} \frac{\partial_{\mathrm{I}} r_{k,11}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,12}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,13}^*}{\partial_{\mathrm{I}}\varphi_k} \\ \frac{\partial_{\mathrm{I}} r_{k,21}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,22}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,23}^*}{\partial_{\mathrm{I}}\varphi_k} \\ \frac{\partial_{\mathrm{I}} r_{k,31}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,32}^*}{\partial_{\mathrm{I}}\varphi_k} & \frac{\partial_{\mathrm{I}} r_{k,33}^*}{\partial_{\mathrm{I}}\varphi_k} \end{pmatrix} = \begin{pmatrix} \mathrm{RP}_{11} & \mathrm{RP}_{12} & \mathrm{RP}_{13} \\ \mathrm{RP}_{21} & \mathrm{RP}_{22} & \mathrm{RP}_{23} \\ \mathrm{RP}_{31} & \mathrm{RP}_{32} & \mathrm{RP}_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} -\sin{}_{\mathrm{I}}\varphi_k \cdot \cos{}_{\mathrm{I}}\kappa_k & -\sin{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\omega_k \cdot \cos{}_{\mathrm{I}}\kappa_k & \cos{}_{\mathrm{I}}\varphi_k \cdot \cos{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\omega_k \\ -\sin{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\kappa_k & \sin{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\omega_k & \cos{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\omega_k \\ -\cos{}_{\mathrm{I}}\varphi_k & -\sin{}_{\mathrm{I}}\omega_k \cdot \sin{}_{\mathrm{I}}\varphi_k & -\sin{}_{\mathrm{I}}\varphi_k \cdot \cos{}_{\mathrm{I}}\omega_k \end{pmatrix}$$

$$\frac{\partial_{\mathrm{I}}\boldsymbol{R}_k^{-1}}{\partial_{\mathrm{I}}\kappa_k} = \begin{pmatrix} \frac{\partial_{\mathrm{I}} r_{k,11}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,12}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,13}^*}{\partial_{\mathrm{I}}\kappa_k} \\ \frac{\partial_{\mathrm{I}} r_{k,21}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,22}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,23}^*}{\partial_{\mathrm{I}}\kappa_k} \\ \frac{\partial_{\mathrm{I}} r_{k,31}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,32}^*}{\partial_{\mathrm{I}}\kappa_k} & \frac{\partial_{\mathrm{I}} r_{k,33}^*}{\partial_{\mathrm{I}}\kappa_k} \end{pmatrix} = \begin{pmatrix} \mathrm{RK}_{11} & \mathrm{RK}_{12} & \mathrm{RK}_{13} \\ \mathrm{RK}_{21} & \mathrm{RK}_{22} & \mathrm{RK}_{23} \\ \mathrm{RK}_{31} & \mathrm{RK}_{32} & \mathrm{RK}_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} -\sin{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\varphi_k & -\cos{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\omega_k - \sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\omega_k & -\sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \,.\cos{}_{\mathrm{I}}\omega_k \\ \cos{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\varphi_k & -\sin{}_{\mathrm{I}}\kappa_k \cdot \cos{}_{\mathrm{I}}\omega_k + \cos{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \cdot \sin{}_{\mathrm{I}}\omega_k & \sin{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\omega_k + \cos{}_{\mathrm{I}}\kappa_k \cdot \sin{}_{\mathrm{I}}\varphi_k \cdot \cos{}_{\mathrm{I}}\omega_k \\ 0 & 0 & 0 \end{pmatrix}$$

Giving the following partial derivatives,

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}\omega_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,k-1}^* \cdot \begin{bmatrix} \mathrm{RO}_{11} & \mathrm{RO}_{12} & \mathrm{RO}_{13} & -(\mathrm{RO}_{11} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RO}_{12} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RO}_{13} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RO}_{21} & \mathrm{RO}_{22} & \mathrm{RO}_{23} & -(\mathrm{RO}_{21} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RO}_{22} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RO}_{23} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RO}_{31} & \mathrm{RO}_{32} & \mathrm{RO}_{33} & -(\mathrm{RO}_{31} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RO}_{32} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RO}_{33} \cdot {}_{\mathrm{I}}t_{k,3}) \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}_{k+1,m}^* \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}\varphi_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}_{0,k-1}^* \cdot \begin{bmatrix} \mathrm{RP}_{11} & \mathrm{RP}_{12} & \mathrm{RP}_{13} & -(\mathrm{RP}_{11} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RP}_{12} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RP}_{13} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RP}_{21} & \mathrm{RP}_{22} & \mathrm{RP}_{23} & -(\mathrm{RP}_{21} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RP}_{22} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RP}_{23} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RP}_{31} & \mathrm{RP}_{32} & \mathrm{RP}_{33} & -(\mathrm{RP}_{31} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RP}_{32} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RP}_{33} \cdot {}_{\mathrm{I}}t_{k,3}) \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}_{k+1,m}^* \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}\kappa_k} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot \begin{bmatrix} \mathrm{RK}_{11} & \mathrm{RK}_{12} & \mathrm{RK}_{13} & -(\mathrm{RK}_{11} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RK}_{12} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RK}_{13} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RK}_{21} & \mathrm{RK}_{22} & \mathrm{RK}_{23} & -(\mathrm{RK}_{21} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RK}_{22} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RK}_{23} \cdot {}_{\mathrm{I}}t_{k,3}) \\ \mathrm{RK}_{31} & \mathrm{RK}_{32} & \mathrm{RK}_{33} & -(\mathrm{RK}_{31} \cdot {}_{\mathrm{I}}t_{k,1} + \mathrm{RK}_{32} \cdot {}_{\mathrm{I}}t_{k,2} + \mathrm{RK}_{33} \cdot {}_{\mathrm{I}}t_{k,3}) \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

<u>*Translation partial derivatives:*</u>

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}t_{k,1}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot \begin{bmatrix} 0 & 0 & 0 & -\cos\kappa_i \cdot \cos\varphi_i \\ 0 & 0 & 0 & -\cos\kappa_i \cdot \sin\varphi_i \\ 0 & 0 & 0 & \sin\varphi_i \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}t_{k,2}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot \begin{bmatrix} 0 & 0 & 0 & -(-\sin\kappa_i \cdot \cos\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i) \\ 0 & 0 & 0 & -(\cos\kappa_i \cdot \cos\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \sin\omega_i) \\ 0 & 0 & 0 & -(\cos\varphi_i \cdot \sin\omega_i) \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}t_{k,3}} = L \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot \begin{bmatrix} 0 & 0 & 0 & -(\sin\kappa_i \cdot \sin\omega_i + \cos\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i) \\ 0 & 0 & 0 & -(-\cos\kappa_i \cdot \sin\omega_i + \sin\kappa_i \cdot \sin\varphi_i \cdot \cos\omega_i) \\ 0 & 0 & 0 & -(\cos\varphi_i \cdot \cos\omega_i) \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

<u>*Scale factor partial derivatives:*</u>

$$\frac{\partial \boldsymbol{X}_{LOR2}}{\partial_{\mathrm{I}}l_k} = ({}_{\mathrm{I}}l_0^* \cdot \dots \cdot {}_{\mathrm{I}}l_{k+1}^* \cdot 1 \cdot {}_{\mathrm{I}}l_{k-1}^* \cdot \dots \cdot {}_{\mathrm{I}}l_m^* \cdot {}_{P}l_n \cdot \dots {}_{P}l_0) \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{0,k-1} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot {}_{\mathrm{I}}\boldsymbol{M}^*_{k+1,m} \cdot {}_{\mathrm{P}}\boldsymbol{M}_{n,0} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

1.2.1.5 Covariance matrix transformation

If covariance matrix of a point is given we also need to transform this matrix.

Covariance matrix of a point $\boldsymbol{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$**,** defined in a local object reference, is:

$$\Sigma_{\boldsymbol{X}} = \mathrm{E}[(\boldsymbol{X} - \mathrm{E}(\boldsymbol{X})) \cdot (\boldsymbol{X} - \mathrm{E}(\boldsymbol{X}))^T] = \begin{pmatrix} \sigma_x^2 & \sigma_{x,y} & \sigma_{x,z} \\ \sigma_{x,y} & \sigma_y^2 & \sigma_{y,z} \\ \sigma_{x,z} & \sigma_{y,z} & \sigma_z^2 \end{pmatrix} \tag{1.2}$$

Transformed covariance matrix of the point in CCS must have form:

$$\Sigma_{\boldsymbol{X}_{CCS}} = \mathrm{E}[(\boldsymbol{X}_{\boldsymbol{CCS}} - \mathrm{E}(\boldsymbol{X}_{\boldsymbol{CCS}})) \cdot (\boldsymbol{X}_{\boldsymbol{CCS}} - \mathrm{E}(\boldsymbol{X}_{\boldsymbol{CCS}}))^T]$$

Translation part of the transformation does not affect the covariance matrix, it is affected only by rotation and scaling.

Rotation part of the transformation, extracted from (1.1):

$$\boldsymbol{M}^R = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}$$

Scale factor, extracted from (1.1):

$$l = \frac{1}{M_{44}}$$

The transformation can be then expressed (without translation part):

$$\boldsymbol{X}_{CCS} = (l \cdot \boldsymbol{M}^R) \cdot \boldsymbol{X} \quad (1.3)$$

Substituting (1.3) into (1.2) and applying the properties of the expectation, in particular

$$\mathrm{E}(l \cdot \boldsymbol{M}^R \cdot \boldsymbol{X}) = l \cdot \boldsymbol{M}^R \cdot \mathrm{E}(\boldsymbol{X})$$

We immediately get

$$\Sigma_{\boldsymbol{X}_{CCS}} = (l \cdot \boldsymbol{M}^R) \cdot \mathrm{E}\left[\left(\boldsymbol{X} - \mathrm{E}(\boldsymbol{X})\right) \cdot \left(\boldsymbol{X} - \mathrm{E}(\boldsymbol{X})\right)^T\right] \cdot (l \cdot \boldsymbol{M}^R)^T$$

And finally, we get the relation between $\Sigma_{\boldsymbol{X}}$ and$\Sigma_{\boldsymbol{X}_{CCS}}$:

$$\Sigma_{\boldsymbol{X}_{CCS}} = (l \cdot \boldsymbol{M}^R) \cdot \Sigma_{\boldsymbol{X}} \cdot (l \cdot \boldsymbol{M}^R)^T \quad (1.4)$$

1.2.2 Transformation from the CERN Coordinate System (CCS) to Instrument Local Astronomical

Transformation chain to transform a point form the CCS to Instrument local astronomical includes three transformations:

1. **CCS -> CGRF**, $(x_i, y_i, z_i)^{CCS} \to (x_i, y_i, z_i)^{CGRF}$, via $\boldsymbol{A}^*$which requires geodetic ellipsoidal coordinates of the initial point of the CCS ($\boldsymbol{P_0}$) -- φ_0, λ_0, its azimuth A_Z and its false and geodetic Cartesian coordinates.
2. **CGRF -> Local Geodetic System**, $(x_i, y_i, z_i)^{CGRF} \to (x_i, y_i, z_i)^{LG}$, via $\boldsymbol{B}_i^*$ which requires $(x_i, y_i, z_i)^{CGRF}$and the ellipsoidal coordinates $\varphi_i, \lambda_i$of this point in the CGRF**.**
3. **Local Geodetic -> Local Astronomical**, $(x_i, y_i, z_i)^{LG} \to (x_i, y_i, z_i)^{LA}$, via $\boldsymbol{C}_i^*$ which requires angles ξ_i, η_i and $\Delta A_i(\eta_i, \varphi_i)$.

<u>Prerequisites:</u>

- $\boldsymbol{P_0}$ — is a principal, or initial, point of the CCS, located at the centre of the PS accelerator (the origin is by definition the centre of the line from P2 to P1).
  <u>False coordinates</u>, at the $\boldsymbol{P_0}$, in meters are defined to be:

$$\boldsymbol{X}_{P_0} = \begin{pmatrix} 2000.00000 \\ 2097.79265 \\ 2433.66000 \end{pmatrix} \quad (2.1)$$

The geodetic ellipsoidal coordinates $(\varphi_0, \lambda_0, h_0)$ of $\boldsymbol{P_0}$ are defined to be:

$$\begin{pmatrix}\varphi_0\\ \lambda_0\\ h_0\end{pmatrix} = \begin{pmatrix}51.3692\text{ gon}\\ 6.72124\text{ gon}\\ 433.65921\text{ metres}\end{pmatrix} \tag{2.2}$$

The azimuth of the CCS Y-axis is defined to be:

$$A_Z = 37.77864\text{ gon} \tag{2.3}$$

- GRS80 reference ellipsoid is defined:

Semi-major axis: $a_{GRS80} = 6\,378\,137.0\text{ m}$ (2.4)

Semi-minor axis: $b_{GRS80} \approx 6\,356\,752.314\,140\text{ m}$

Eccentricity: $e^2 = \frac{a^2-b^2}{a^2}$

Second eccentricity: $e' = \sqrt{\frac{e^2}{1-e^2}}$

1.2.2.1 CCS to CERN Geodetic Syxtem (CGRF)

The CCS is a *local geodetic system* and the CGRF is a *geodetic reference system*. Based on (Jones, 2000), the transformation of a point in CCS ($\mathbf{X}_{\mathrm{CCS,cc}}$) to CGRF ($\mathbf{X}_{\mathrm{CGRF,cc}}$) can be expressed (in Cartesian coordinates) as :

$$\boldsymbol{X}_{CGRF,cc} = \boldsymbol{A}_{cc}^{*} \cdot \left(\boldsymbol{X}_{CCS,cc} - \boldsymbol{X}_{P_0,cc}\right) + \boldsymbol{X}_{cc}^{GC}, \tag{2.5}$$

Where:

$\boldsymbol{A}_{cc}^{*} = R_z(\pi - \lambda_0) \cdot R_y\left(\frac{\pi}{2} - \varphi_0\right) \cdot R_z\left(\frac{\pi}{2}\right) \cdot R_z(-A_Z)$ – is a rotation matrix. The parameters φ_0, λ_0 and the azimuth A_Z are well-defined (2.2, 2.3) and therefore it can be seen as constant:

$$\boldsymbol{A}_{cc}^{*} = \tag{2.6}$$

$$\begin{pmatrix} -\cos(-A_z)\cdot\sin\lambda_0 - \sin(-A_z)\cdot\cos\lambda_0\cdot\sin\varphi_0 & -\cos\lambda_0\cdot\sin\varphi_0\cdot\cos(-A_z) + \sin\lambda_0\cdot\sin(-A_z) & \cos\lambda_0\cdot\cos\varphi_0 \\ \cos(-A_z)\cdot\cos\lambda_0 - \sin(-A_z)\cdot\sin\lambda_0\cdot\sin\varphi_0 & -\sin\lambda_0\cdot\sin\varphi_0\cdot\cos(-A_z) - \cos\lambda_0\cdot\sin(-A_z) & \sin\lambda_0\cdot\cos\varphi_0 \\ \sin(-A_z)\cdot\cos\varphi_0 & \cos(-A_z)\cdot\cos\varphi_0 & \sin\varphi_0 \end{pmatrix} =$$

$$= \begin{pmatrix} 0.3142167651 & -0.6542800175 & 0.6878847891 \\ 0.8669698182 & 0.4930005365 & 0.0728958522 \\ -0.3868218696 & 0.5734702517 & 0.7221500616 \end{pmatrix} =$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix}$$

- $\boldsymbol{X}_{P_0,cc}$—false coordinates at the $\boldsymbol{P_0}$, in metres (2.1).
- $\boldsymbol{X}_{cc}^{GC}$—geodetic Cartesian coordinates of the $\boldsymbol{P_0}$. Transformation of this point from the geodetic ellipsoidal coordinates to the geodetic Cartesian coordinates (using GRS80 reference ellipsoid (2.4), where ϑ Is a prime vertical radius of curvature, at latitude φ_0:

$$\vartheta_{GRS80,P0} = \frac{{a_{GRS80}}^2}{(1 - e^2 \cdot \sin^2 \varphi_0)^{\frac{1}{2}}} \approx 6389299.669619136$$

Is:

$$\boldsymbol{X}_{cc}^{GC} = \begin{pmatrix} (\vartheta + h) \cdot \cos \varphi \cdot \cos \lambda \\ (\vartheta + h) \cdot \cos \varphi \cdot \sin \lambda \\ (\vartheta \cdot (1 - e^2) + h) \cdot \sin \varphi \end{pmatrix} =$$

$$\left| \begin{matrix} \varphi := \varphi_0, \lambda := \lambda_0, h := h_0 \\ \vartheta := \vartheta_{GRS80,P0} \\ a := a_{GRS80}, b := b_{GRS80} \end{matrix} \right| = \begin{pmatrix} 4395400.363781726 \\ 465785.056735627 \\ 4583458.226013717 \end{pmatrix} = \begin{pmatrix} x^{GC} \\ y^{GC} \\ z^{GC} \end{pmatrix}$$

Equation (2.5) can be rewritten in homogeneous coordinates:

$$\boldsymbol{X}_{CGRF} = \boldsymbol{A}^* \cdot \boldsymbol{X}_{CCS} \tag{2.7}$$

$$\frac{\partial \boldsymbol{X}_{CGRF}}{\partial *} = \boldsymbol{A}^* \cdot \frac{\partial \boldsymbol{X}_{CCS}}{\partial *}$$

Symbol * can stand for: $x_0; y_0;\ z_0;\ \omega_j; \varphi_j; k_{j;}\ t_{j,1}; t_{j,2}; t_{j,3}; l_j$.

Where:

$$\boldsymbol{A}^* = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & t_{A,1} + x^{GC} \\ a_{2,1} & a_{2,2} & a_{2,3} & t_{A,2} + y^{GC} \\ a_{3,1} & a_{3,2} & a_{3,3} & t_{A,3} + z^{GC} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} t_{A,1} \\ t_{A,2} \\ t_{A,3} \end{pmatrix} = \boldsymbol{A}_{cc}^* \cdot (-\boldsymbol{X}_{P_0,cc}) = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix} \cdot \begin{pmatrix} -2000.00000 \\ -2097.79265 \\ -2433.66000 \end{pmatrix}$$

1.2.2.2 CGRF to Local Geodetic System

Is a transformation from a *geodetic reference system* to a *local geodetic system* which requires ellipsoidal coordinates φ_i, λ_i of the origin of local geodetic system into which we transform (typically coordinates of the instrument) $\boldsymbol{X}_{orig,CGRF}$. These ellipsoidal coordinates can be obtained from the geodetic Cartesian coordinates of the point either via iterative algorithm described in ((Jones, 2000) , page 8) or with use of closed form approximation:

$$\gamma = \arctan\left(\frac{Z \cdot a}{\sqrt{X^2 + Y^2} \cdot b}\right) \tag{2.8}$$

$$\lambda = \arctan\left(\frac{Y}{X}\right)$$

$$\varphi = \arctan\left(\frac{Z + e'^2 \cdot b \cdot \sin^3 \gamma}{\sqrt{X^2 + Y^2} - e^2 \cdot a \cdot \cos^3 \gamma}\right)$$

$$h = \frac{\sqrt{X^2 + Y^2}}{\cos\varphi} - \vartheta$$

Where a, b, e'define reference ellipsoid, we use (2.4), ϑ is a prime vertical radius of curvature and X, Y, Z are geodetic Cartesian coordinates of the origin in CGRF.
Test of differences between these two methods was conducted and the differences were less than 10^{-8}m (in Cartesian coordinates), (Zwiener, 2011).

Although, $\varphi_i, \lambda_i$are functions of the unknown elements of $\mathbf{X}_{\mathrm{CCS,cc}} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, they will be treated as constants. Still, these values should be recalculated in each iteration of the adjustment process.

For h_i, we have:

$$\frac{\partial\, h_i}{\partial\, *} = \frac{X_{CGRF}}{\cos\varphi_i \cdot \sqrt{{X_{CGRF}}^2 + {Y_{CGRF}}^2}} \cdot \frac{\partial X_{CGRF}}{\partial\, *} + \frac{Y_{CGRF}}{\cos\varphi_i \cdot \sqrt{{X_{CGRF}}^2 + {Y_{CGRF}}^2}} \cdot \frac{\partial Y_{CGRF}}{\partial\, *}$$

Symbol * can stand for: $x_0; y_0;\ z_0;\ \omega_j; \varphi_j; \kappa_{j;}\, t_{j,1}; t_{j,2}; t_{j,3}; l_j$.

The transformation can be expressed (in Cartesian coordinates) as:

$$\boldsymbol{X}_{LG,cc} = \boldsymbol{B}^*_{\boldsymbol{i,cc}} \cdot (\boldsymbol{X}_{CGRF,cc} - \boldsymbol{X}^{GC}_{orig,cc})$$

Where

$$\boldsymbol{B}^*_{\boldsymbol{i,cc}} = (R_z(\pi - \lambda_i) \cdot R_y\left(\frac{\pi}{2} - \varphi_i\right) \cdot R_z\left(\frac{\pi}{2}\right))^T = \begin{pmatrix} -\sin\lambda_i & \cos\lambda_i & 0 \\ -\cos\lambda_i \cdot \sin\varphi_i & -\sin\lambda_i \cdot \sin\varphi_i & \cos\varphi_i \\ \cos\lambda_i \cdot \cos\varphi_i & \cos\varphi_i \cdot \sin\lambda_i & \sin\varphi_i \end{pmatrix}$$
$$= \begin{pmatrix} b_{i,11} & b_{i,12} & b_{i,13} \\ b_{i,21} & b_{i,22} & b_{i,23} \\ b_{i,31} & b_{i,32} & b_{i,33} \end{pmatrix}$$

In homogenous coordinates we get:

$$\boldsymbol{X}_{LG} = \boldsymbol{B}^*_{\boldsymbol{i}} \cdot \boldsymbol{X}_{CGRF}$$

Where

$$\boldsymbol{B}^*_{\boldsymbol{i}} = \begin{bmatrix} b_{i,11} & b_{i,12} & b_{i,13} & t_{B,1} \\ b_{i,21} & b_{i,22} & b_{i,23} & t_{B,2} \\ b_{i,31} & b_{i,32} & b_{i,33} & t_{B,3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{pmatrix} t_{B,1} \\ t_{B,2} \\ t_{B,3} \end{pmatrix} = \boldsymbol{B}^*_{i,cc} \cdot (-\boldsymbol{X}^{GC}_{cc}) = -\begin{pmatrix} b_{i,11} \cdot x^{GC}_{orig} + b_{i,12} \cdot y^{GC}_{orig} \\ b_{i,21} \cdot x^{GC}_{orig} + b_{i,22} \cdot y^{GC}_{orig} + b_{i,23} \cdot z^{GC}_{orig} \\ b_{i,31} \cdot x^{GC}_{orig} + b_{i,32} \cdot y^{GC}_{orig} + b_{i,33} \cdot z^{GC}_{orig} \end{pmatrix}$$

The transformation from the CCS to the Instrument System Geodetic is then:

$$\boldsymbol{X}_{LG} = \boldsymbol{B}^*_{\boldsymbol{i}} \cdot \boldsymbol{A}^* \cdot \boldsymbol{X}_{CCS}$$

$$\frac{\partial \boldsymbol{X}_{LG}}{\partial *} = \boldsymbol{B}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{A}^{*} \cdot \frac{\partial \boldsymbol{X}_{CCS}}{\partial *}$$

Symbol * can stand for: $x_0; y_0;\ z_0;\ \omega_j; \varphi_j; \kappa_{j;}\ t_{j,1}; t_{j,2}; t_{j,3}; l_j$.

1.2.2.3 Local Geodetic to Local Astronomical

This is a transformation from a *local geodetic system* to a *local astronomical system* (see (Jones, 2000)). It requires angles $\xi_i,\ \eta_i$ of the local system origin which can be obtained from the grid and $\Delta A_i =\ \eta_i \cdot \tan\varphi_i$.

Rotation part of this transformation is:

$$\boldsymbol{C}_{\boldsymbol{i,cc}}^{*} = \begin{pmatrix} 1 & \eta_i \cdot \tan\varphi_i & -\eta_i \\ -\eta_i \cdot \tan\varphi_i & 1 & -\xi_i \\ \eta_i & \xi_i & 1 \end{pmatrix}$$

In homogenous coordinates we get:

$$\boldsymbol{X}_{ILA} = \boldsymbol{C}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{X}_{LG}$$

Where

$$\boldsymbol{C}_{\boldsymbol{i}}^{*} = \begin{pmatrix} 1 & \eta_i \cdot \tan\varphi_i & -\eta_i & 0 \\ -\eta_i \cdot \tan\varphi_i & 1 & -\xi_i & 0 \\ \eta_i & \xi_i & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

1.2.2.4 Conclusion

Based on concepts described above we finally conclude with formula for a point transformation from the CCS to a local instrument astronomical:

$$\boldsymbol{X}_{LA} = \boldsymbol{C}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{B}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{A}^{*} \cdot \boldsymbol{X}_{CCS}$$

Partial derivatives:

$$\frac{\partial \boldsymbol{X}_{LA}}{\partial *} = \boldsymbol{C}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{B}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{A}^{*} \cdot \frac{\partial \boldsymbol{X}_{CCS}}{\partial *}$$

Symbol * can stand for: $x_0; y_0;\ z_0;\ \omega_j; \varphi_j; \kappa_{j;}\ t_{j,1}; t_{j,2}; t_{j,3}; l_j$.

For a transformation from a local system we get:

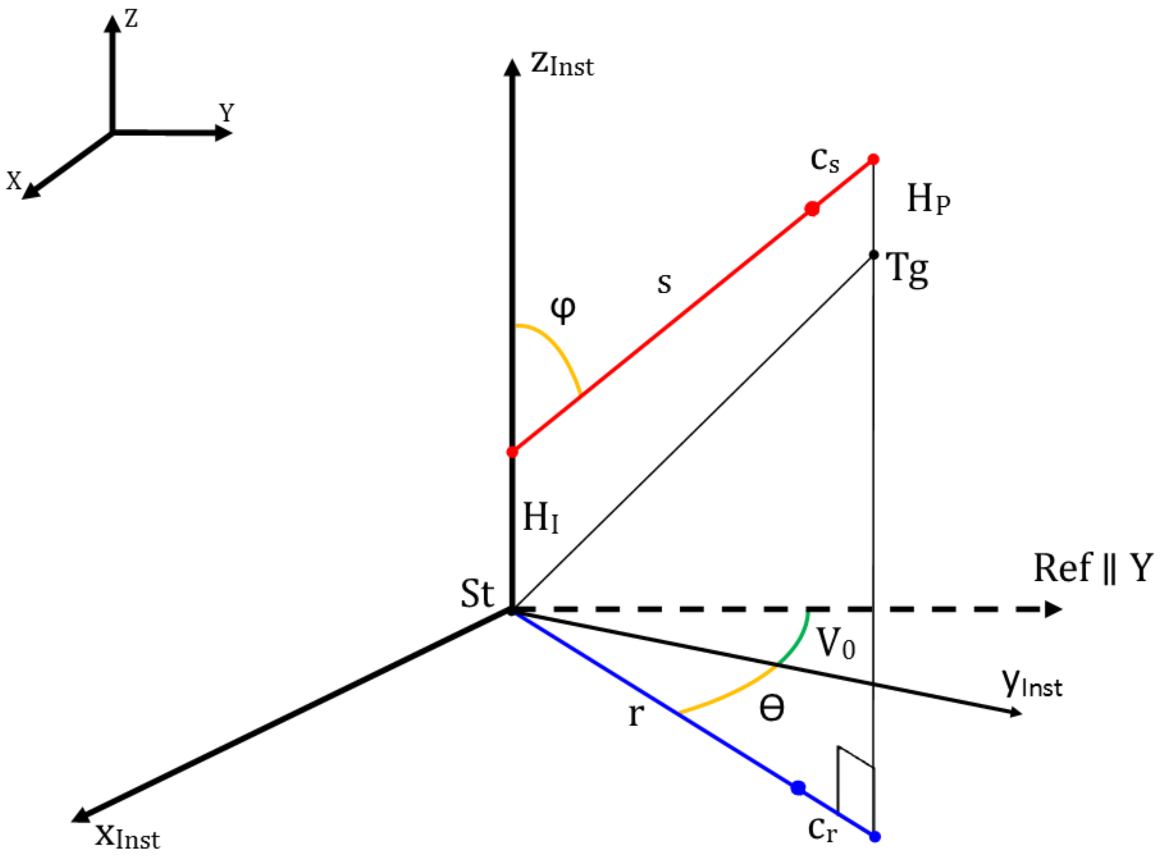
$$\boldsymbol{X}_{ILA} = \boldsymbol{C}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{B}_{\boldsymbol{i}}^{*} \cdot \boldsymbol{A}^{*} \cdot \boldsymbol{M} \cdot \boldsymbol{x}_0^{\boldsymbol{h}}$$

Covariance matrix transformation:

Applying the same principle for transforming *covariance matrix*, as described in the section Point Transformations, and noting that none of the three transformations described above includes scale factor, we get a relationship between the covariance matrix in a local system $\boldsymbol{\Sigma}$ and covariance matrix in Instrument Local Astronomical$\boldsymbol{\Sigma}_{ILA}$:

$$\boldsymbol{\Sigma}_{ILA} = (\boldsymbol{C}^{*}_{\boldsymbol{i,cc}} \cdot \boldsymbol{B}^{*}_{\boldsymbol{i,cc}} \cdot \boldsymbol{A}^{*}_{cc} \cdot (l \cdot \boldsymbol{M}^{R})) \cdot \boldsymbol{\Sigma} \cdot \left(\boldsymbol{C}^{*}_{\boldsymbol{i,cc}} \cdot \boldsymbol{B}^{*}_{\boldsymbol{i,cc}} \cdot \boldsymbol{A}^{*}_{cc} \cdot (l \cdot \boldsymbol{M}^{R})\right)^{T}$$

2. POLAR INSTRUMENT OBSERVATION MODELS



2.1 Polar measurement (PLR3D) for a TSTN

This measurement is made in the Instrument Local Astronomical System (ILA).
If Z_{Tg} is not vertical, H_P has to be zero.

2.1.1 Mathematical Model

$$\begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varphi & \sin\varphi \\ 0 & -\sin\varphi & \cos\varphi \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ s + c_s \end{pmatrix}$$
$$-\begin{pmatrix} \cos V_0 & -\sin V_0 & 0 \\ \sin V_0 & \cos V_0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} \cos Ry & 0 & \sin Ry \\ 0 & 1 & 0 \\ -sinRy & 0 & \cos Ry \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos Rx & -\sin Rx \\ 0 & \sin Rx & \cos Rx \end{pmatrix}\begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St} + H_I) \end{pmatrix} = 0$$

$$F = (s + c_s).\begin{pmatrix} \sin\theta \cdot \sin\varphi \\ \cos\theta \cdot \sin\varphi \\ \cos\varphi \end{pmatrix} - A\begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St} + H_I) \end{pmatrix} = 0$$

Where,

$$\boldsymbol{A} = \begin{pmatrix} \cos V_0 \,.\cos Ry & -\sin V_0 \,.\cos Rx + \sin R_x \,.\cos V_0 \,.\sin Ry & \sin V_0 \,.\sin Rx + \cos R_x \,.\cos V_0 \,.\sin Ry \\ \sin V_0 \,.\cos Ry & \cos V_0 \,.\cos Rx + \sin R_x \,.\sin V_0 \,.\sin Ry & -\cos V_0 \,.\sin Rx + \cos R_x \,.\sin V_0 \,.\sin Ry \\ -\sin Ry & \sin Rx \,.\cos Ry & \cos Rx \,.\cos Ry \end{pmatrix}$$

2.1.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial *} = \boldsymbol{A} \cdot \begin{pmatrix} \frac{\partial X_{St}}{\partial *} \\ \frac{\partial Y_{St}}{\partial *} \\ \frac{\partial Z_{St}}{\partial *} \end{pmatrix}$$

Where:

Symbol * can stand for: $x_{0,St}; y_{0,St};\ z_{0,St};\ \omega_{j,St}; \varphi_{j,St}; \kappa_{j,St;}\ t_{j,1,St}; t_{j,2,St}; t_{j,3,St}; l_{j,St}.$ (2.9)

For the Target coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial *} = -\boldsymbol{A} \cdot \begin{pmatrix} \frac{\partial X_{Tg}}{\partial *} \\ \frac{\partial Y_{Tg}}{\partial *} \\ \frac{\partial Z_{St}}{\partial *} \end{pmatrix}$$

Where:

* can stand for: $x_{0,Tg};\ y_{0,Tg};\ z_{0,Tg};\ \omega_{j,Tg}; \varphi_{j,Tg}; \kappa_{j,Tg}; t_{j,1,Tg}; t_{j,2,Tg}; t_{j,3,Tg}; l_{j,Tg}.$ (2.10)

For the additional parameters:

$$\frac{\partial F}{\partial V_0} = -\begin{pmatrix} -\sin V_0 \,.\cos Ry & -\cos V_0 \,.\cos Rx - \sin R_x \,.\sin V_0 \,.\sin Ry & \cos V_0 \,.\sin Rx - \cos R_x \,.\sin V_0 \,.\sin Ry \\ \cos V_0 \,.\cos Ry & -\sin V_0 \,.\cos Rx + \sin R_x \,.\cos V_0 \,.\sin Ry & \sin V_0 \,.\sin Rx + \cos R_x \,.\cos V_0 \,.\sin Ry \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St} + H_I) \end{pmatrix}$$

$$\frac{\partial F}{\partial c_s} = \begin{pmatrix} \sin\theta \cdot \sin\varphi \\ \cos\theta \cdot \sin\varphi \\ \cos\varphi \end{pmatrix}$$

$$\frac{\partial F}{\partial Rx} = -\begin{pmatrix} 0 & \sin V_0 . \sin Rx + \cos R_x . \cos V_0 . \sin Ry & \sin V_0 . \cos Rx - \sin R_x . \cos V_0 . \sin Ry \\ 0 & -\cos V_0 . sin + \cos R_x . \sin V_0 . \sin Ry & -\cos V_0 . \cos Rx - \sin R_x . \sin V_0 . \sin Ry \\ 0 & \cos Rx . \cos Ry & -\sin Rx . \cos Ry \end{pmatrix} \begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St} + H_I) \end{pmatrix}$$

$$\frac{\partial F}{\partial Ry} = -\begin{pmatrix} -\cos V_0 . \sin Ry & \sin R_x . \cos V_0 . \cos Ry & \cos R_x . \cos V_0 . \cos Ry \\ -\sin V_0 . \sin Ry & \sin R_x . \sin V_0 . \cos Ry & \cos R_x . \sin V_0 . \cos Ry \\ -\cos Ry & -\sin Rx . \sin Ry & -\cos Rx . \sin Ry \end{pmatrix} \begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St} + H_I) \end{pmatrix}$$

For the observed measurements:

$$\frac{\partial F}{\partial \theta} = (s + c_s). \begin{pmatrix} \cos \theta \cdot \sin \varphi \\ -\sin \theta \cdot \sin \varphi \\ 0 \end{pmatrix}$$

$$\frac{\partial F}{\partial \varphi} = (s + c_s). \begin{pmatrix} \sin \theta \cdot \cos \varphi \\ \cos \theta \cdot \cos \varphi \\ -\sin \varphi \end{pmatrix}$$

$$\frac{\partial F}{\partial s} = \begin{pmatrix} \sin \theta \cdot \sin \varphi \\ \cos \theta \cdot \sin \varphi \\ \cos \varphi \end{pmatrix}$$

2.1.3 Stochastic model

For all the document, the stochastic model is calculated with the law of standard deviation propagation:

$$\sigma_l^2 = \sum_i (\frac{\partial l}{\partial x_i})^2 \sigma_i^2.$$

Where:
x_i -- Parameters with their standard deviation σ_i.
σ_l -- Standard deviation of the observation.

Then we suppose that:
σ_m Is the standard deviation relative to the instrument and material (for example ppm for a distance metre).
σ_{CSt} Is the standard deviation relative to the centring of the instrument (in XY plan, both part are equal).
σ_{CTg} Is the standard deviation relative to the centring of the target (in XY plan, both part are equal).

For PLR3D measurement, $\sigma_\theta{}^2, \sigma_\varphi{}^2$ *and* $\sigma_s{}^2$ are described below in ANGL, ZEND and DIST.

2.1.4 Objects

Adjustable needed:

- **Point** – for the *Station* and the *Target*
- **Angle** – for the V_0
- **Scalar --** Instrument height H_I

Simple objects (**not adjustable**):

- **Angle** – for the θ and φ
- **Scalar** – for the s
- **Scalar** – target height H_P
- **Scalar –** for the standard deviations

2.1 Horizontal direction (ANGL)

This measurement is made in the Modified Local Astronomical System (MLA).

2.1.1 Mathematical Model

$$\theta = \tan^{-1}\left(\frac{X_{Tg} - X_{St}}{Y_{Tg} - Y_{St}}\right) - V_0$$

2.1.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial\theta}{\partial *} = -\frac{\left(Y_{Tg} - Y_{St}\right)}{D^2} \cdot \frac{\partial X_{St}}{\partial *} + \frac{\left(X_{Tg} - X_{St}\right)}{D^2} \cdot \frac{\partial Y_{St}}{\partial *}$$

Where

$$D^2 = \left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2$$

Where (2.9).
For the Target coordinates:

$$\frac{\partial\theta}{\partial *} = \frac{\left(Y_{Tg} - Y_{St}\right)}{D^2} \cdot \frac{\partial X_{Tg}}{\partial *} + \frac{\left(X_{St} - X_{Tg}\right)}{D^2} \cdot \frac{\partial Y_{Tg}}{\partial *}$$

Where (2.10).
For the additional parameters:

$$\frac{\partial\theta}{\partial V_0} = -1$$

2.1.3 Stochastic model

$$\sigma_\theta{}^2 = \sigma_m{}^2 + \frac{1}{D^2}\left(\sigma_{CTg}{}^2 + \sigma_{CSt}{}^2\right)$$

And σ_m represents the standard deviation of the measured vertical angle.

2.1.4 Objects

Adjustable needed:

- **Point** – for the *Target* and *Station*
- **Angle** – for the V_0

Simple objects (**not adjustable**):

- **Angle** – for the θ
- **Scalar -** for the standard deviations $\sigma_m, \sigma_{V0}, \sigma_{CSt}, \sigma_{CTg}$

2.2 Zenith distance (ZENI / ZENH / XXXX)

This measurement is made in the Modified Local Astronomical System (MLA).
If Z_{Tg} is not vertical, H_P has to be zero.

2.2.1 Mathematical Model

$$\varphi = \cos^{-1}\left(\frac{\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)}{D_{XYZ}}\right)$$

Where

$$D_{XYZ} = \sqrt{\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2 + \left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]^2}$$

2.2.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial\varphi}{\partial *} = \left(\frac{-\left(X_{Tg} - X_{St}\right)\cdot\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]}{(D_{XYZ})^3\cdot\sin\varphi}\right)\cdot\frac{\partial X_{St}}{\partial *} + \left(\frac{-\left(Y_{Tg} - Y_{St}\right)\cdot\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]}{(D_{XYZ})^3\cdot\sin\varphi}\right)\cdot\frac{\partial Y_{St}}{\partial *} + \left(\frac{1}{D_{XYZ}\cdot\sin\varphi} - \frac{\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]^2}{(D_{XYZ})^3\cdot\sin\varphi}\right)\cdot\frac{\partial Z_{St}}{\partial *}$$

Where (2.9).

For the Target coordinates:

$$\frac{\partial\varphi}{\partial *} = \left(\frac{\left(X_{Tg} - X_{St}\right)\cdot\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]}{(D_{XYZ})^3\cdot\sin\varphi}\right)\cdot\frac{\partial X_{Tg}}{\partial *} + \frac{\left(Y_{Tg} - Y_{St}\right)\cdot\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]}{(D_{XYZ})^3\cdot\sin\varphi}\cdot\frac{\partial Y_{Tg}}{\partial *} + \left(-\frac{1}{D_{XYZ}\cdot\sin\varphi} + \frac{\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]^2}{(D_{XYZ})^3\cdot\sin\varphi}\right)\cdot\frac{\partial Z_{Tg}}{\partial *}$$

Where (2.10).

For the additional parameters

$$\frac{\partial\varphi}{\partial H_I} = \frac{1}{D_{XYZ}\cdot\sin\varphi} - \frac{\left[\left(Z_{Tg} + H_P\right) - (Z_{St} + H_I)\right]^2}{(D_{XYZ})^3\cdot\sin\varphi}$$

2.2.3 Stochastic model

$$\sigma_\varphi{}^2 = \sigma_m{}^2 + \frac{\left[\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2\right] \cdot \left[\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)\right]^2}{\left((D_{XYZ})^3 \cdot \sin\varphi\right)^2}\left(\sigma_{CTg}{}^2 + \sigma_{CSt}{}^2\right)$$
$$+ \left(-\frac{1}{D_{XYZ} \cdot \sin\varphi} + \frac{\left[\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)\right]^2}{(D_{XYZ})^3 \cdot \sin\varphi}\right)^2 \left(\sigma_{Hi}{}^2 + \sigma_{Hp}{}^2\right)$$

And σ_m represents the standard deviation of the measured zenith distance.

2.2.4 Objects

Adjustable needed:

- **Point** – for the *Station* and *Target*
- **Scalar --** Instrument height H_I

Simple objects (**not adjustable**):

- **Scalar --** target height H_P
- **Angle** – for the φ
- **Scalar** – for the standard deviations

2.3 Spatial Distance (DMES / DTHE / XXXX)

This measurement is made either in the Modified Local Astronomical System (MLA) or in the CCS, both cases are possible.
If Z_{Tg} is not vertical, H_P has to be zero.

2.3.1 Mathematical Model

$$s = \sqrt{\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2 + \left[\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)\right]^2} - c_s$$

2.3.2 Partial Derivatives

Let

$$D = \sqrt{\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2 + \left[\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)\right]^2}$$

For the Station coordinates:

$$\frac{\partial s}{\partial *} = \frac{-\left(X_{Tg} - X_{St}\right)}{D} \cdot \frac{\partial X_{St}}{\partial *} + \left(\frac{-\left(Y_{Tg} - Y_{St}\right)}{D} \cdot \frac{\partial Y_{St}}{\partial *}\right) + \left(-\frac{\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)}{D} \cdot \frac{\partial Z_{St}}{\partial *}\right)$$

Where (2.9).

For the Target coordinates:

$$\frac{\partial s}{\partial *} = \frac{\left(X_{Tg} - X_{St}\right)}{D} \cdot \frac{\partial X_{Tg}}{\partial *} + \frac{\left(Y_{Tg} - Y_{St}\right)}{D} \cdot \frac{\partial Y_{Tg}}{\partial *} + \frac{\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)}{D} \cdot \frac{\partial Z_{Tg}}{\partial *}$$

Where (2.10).

For the additional parameters

$$\frac{\partial s}{\partial H_I} = -\frac{\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)}{D}$$

$$\frac{\partial s}{\partial c_s} = -1$$

2.3.3 Stochastic model

$$\sigma_s{}^2 = \sigma_m{}^2 + \left(\frac{\left(Z_{Tg} + H_P\right) - \left(Z_{St} + H_I\right)}{D}\right)^2 \left(\sigma_{Hi}{}^2 + \sigma_{Hp}{}^2\right) + \frac{\left(Y_{Tg} - Y_{St}\right)^2 + \left(X_{St} - X_{Tg}\right)^2}{D^2}\left(\sigma_{CTg}{}^2 + \sigma_{CSt}{}^2\right)$$

And σ_m represents the standard deviation of the measured distance and the ppm error

$$\sigma_m = \sigma_{obs}(mm) + s(km) * \sigma_{ppm}(mm)$$

2.3.4 Objects

Adjustable needed:

- **Point** – for the *Station* and *Target*
- **Scalar** – c_s (specific for a pair *instrument - target*)
- **Scalar –** Instrument height H_I

Simple objects (**not adjustable**):

- **Scalar** – target height H_P
- **Scalar**– for the s
- **Scalar –** for the standard deviations

2.4 Horizontal distance (DHOR)

This measurement is made in the Modified Local Astronomical System (MLA).

2.4.1 Mathematical Model

$$r = \sqrt{\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2} - c_r$$

2.4.2 Partial Derivatives

Let

$$D = \sqrt{\left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2}$$

For the Station coordinates:

$$\frac{\partial r}{\partial *} = \frac{-\left(X_{Tg} - X_{St}\right)}{D} \cdot \frac{\partial X_{St}}{\partial *} + \frac{-\left(Y_{Tg} - Y_{St}\right)}{D} \cdot \frac{\partial Y_{St}}{\partial *}$$

Where (2.9).

For the Target coordinates:

$$\frac{\partial r}{\partial *} = \frac{\left(X_{Tg} - X_{St}\right)}{D} \cdot \frac{\partial X_{Tg}}{\partial *} + \frac{\left(Y_{Tg} - Y_{St}\right)}{D} \cdot \frac{\partial Y_{Tg}}{\partial *}$$

Where (2.10).

For the additional parameters

$$\frac{\partial r}{\partial c_r} = -1$$

2.4.3 Stochastic model

$$\sigma_r{}^2 = \sigma_m{}^2 + \left(\sigma_{CTg}{}^2 + \sigma_{CSt}{}^2\right)$$

And σ_m represents the standard deviation of the measured distance and the ppm error

$$\sigma_m \quad = \sigma_{obs}(mm) \quad + r(km) \quad * \sigma_{ppm}(mm)$$

2.4.4 Objects

Adjustable needed:

- **Point** – for the *Station* and *Target*
- **Scalar --** for the c_r

Simple objects (**not adjustable**):

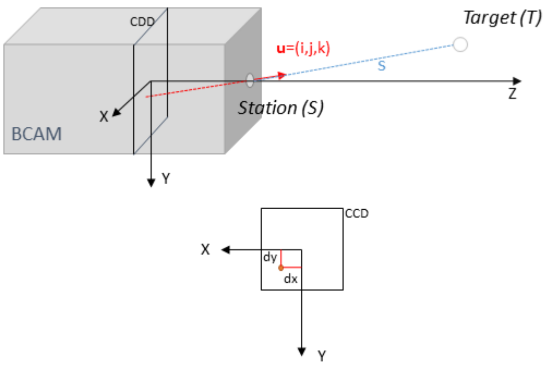
- **Scalar --** for the r
- **Scalar –** for the standard deviations

3. CAMERA INSTRUMENT OBSERVATION MODELS

3.1 Polar measurement (UVD, Unit Vector and Distance) for a CAM

For the camera, the instrument can only rotate freely around the Z axis.
The measurement is described by a unit vector (i j k) + s pointing in a direction of Z-axis, where i, j and s are observations, and k a real to have a unit vector:

- i, is the offset in x between the direction station – target at the target point and the Z axis
- j, is the offset in y between the direction station – target at the target point and the Z axis
- S is the distance station- target



3.1.1 *Mathematical Model*

$$F = s.\begin{pmatrix} i \\ j \\ k \end{pmatrix} - \begin{pmatrix} X_{Tg} - X_{St} \\ Y_{Tg} - Y_{St} \\ (Z_{Tg} + H_P) - (Z_{St}) \end{pmatrix} = 0$$

3.1.2 *Partial Derivatives*

For the Station coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial *} = \begin{pmatrix} \frac{\partial X_{St}}{\partial *} \\ \frac{\partial Y_{St}}{\partial *} \\ \frac{\partial Z_{St}}{\partial *} \end{pmatrix}$$

For the Target coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial *} = -\begin{pmatrix} \frac{\partial X_{Tg}}{\partial *} \\ \frac{\partial Y_{Tg}}{\partial *} \\ \frac{\partial Z_{St}}{\partial *} \end{pmatrix}$$

Where:

$$* \text{ can stand for: } x_{0,Tg};\ y_{0,Tg};\ z_{0,Tg};\ \omega_{j,Tg}; \varphi_{j,Tg}; \kappa_{j,Tg}; t_{j,1,Tg}; t_{j,2,Tg}; t_{j,3,Tg}; l_{j,Tg}. \quad (2.10)$$

For the observed measurements:

$$\frac{\partial F}{\partial i} = s.\begin{pmatrix}1\\0\\0\end{pmatrix}$$

$$\frac{\partial F}{\partial j} = s.\begin{pmatrix}0\\1\\0\end{pmatrix}$$

$$\frac{\partial F}{\partial s} = \begin{pmatrix}i\\ \mathrm{j}\\k\end{pmatrix}$$

3.1.3 Objects

Adjustable needed:

- **Point** – for the *Station* and the *Target*
- **Scalar** – for k

Simple objects (**not adjustable**):

- **Scalar** – for the $i, j \text{ and } s$
- **Scalar** – target height H_P
- **Scalar –** for the standard deviations

3.2 Direction measurement (UVEC, Unit VECtor) for a CAM

3.2.1 Mathematical Model

For the camera, the instrument can only rotate freely around the Z axis. The measurement is described by a unit vector (i j k) pointing in a direction of Z-axis.

3.2.2 Mathematical Model

$$F = \begin{pmatrix}i\\ \mathrm{j}\\k\end{pmatrix} - \frac{1}{\sqrt{\left(X_{Tg}-X_{St}\right)^2+\left(Y_{Tg}-Y_{St}\right)^2+(Z_{Tg}-Z_{St})^2}}\begin{pmatrix}X_{Tg}-X_{St}\\Y_{Tg}-Y_{St}\\Z_{Tg}-Z_{St}\end{pmatrix} = 0$$

3.2.3 Partial Derivatives

Let $D = \sqrt{\left(X_{Tg}-X_{St}\right)^2+\left(Y_{Tg}-Y_{St}\right)^2+(Z_{Tg}-Z_{St})^2}$

For the Target coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial X_{Tg}} = -\frac{1}{\mathrm{D}^3} * \begin{pmatrix} \mathrm{D}^2 + \left(X_{Tg} - X_{St}\right)^2 \\ \left(X_{Tg} - X_{St}\right)^2 \\ \left(X_{Tg} - X_{St}\right)^2 \end{pmatrix}$$

$$\frac{\partial \boldsymbol{F}}{\partial Y_{Tg}} = -\frac{1}{\mathrm{D}^3} * \begin{pmatrix} \left(Y_{Tg} - Y_{St}\right)^2 \\ \mathrm{D}^2 + \left(Y_{Tg} - Y_{St}\right)^2 \\ \left(Y_{Tg} - Y_{St}\right)^2 \end{pmatrix}$$

$$\frac{\partial \boldsymbol{F}}{\partial Z_{Tg}} = -\frac{1}{\mathrm{D}^3} * \begin{pmatrix} \left(Z_{Tg} - Z_{St}\right)^2 \\ \left(Z_{Tg} - Z_{St}\right)^2 \\ \mathrm{D}^2 + \left(Z_{Tg} - Z_{St}\right)^2 \end{pmatrix}$$

For the Station coordinates:

$$\frac{\partial \boldsymbol{F}}{\partial *_{St}} = -\frac{\partial \boldsymbol{F}}{\partial *_{Tg}}$$

For the observed measurements:

$$\frac{\partial F}{\partial i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\frac{\partial F}{\partial j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\frac{\partial F}{\partial k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

3.2.4 Objects

Adjustable needed:

- **Point** – for the *Station* and the *Target*

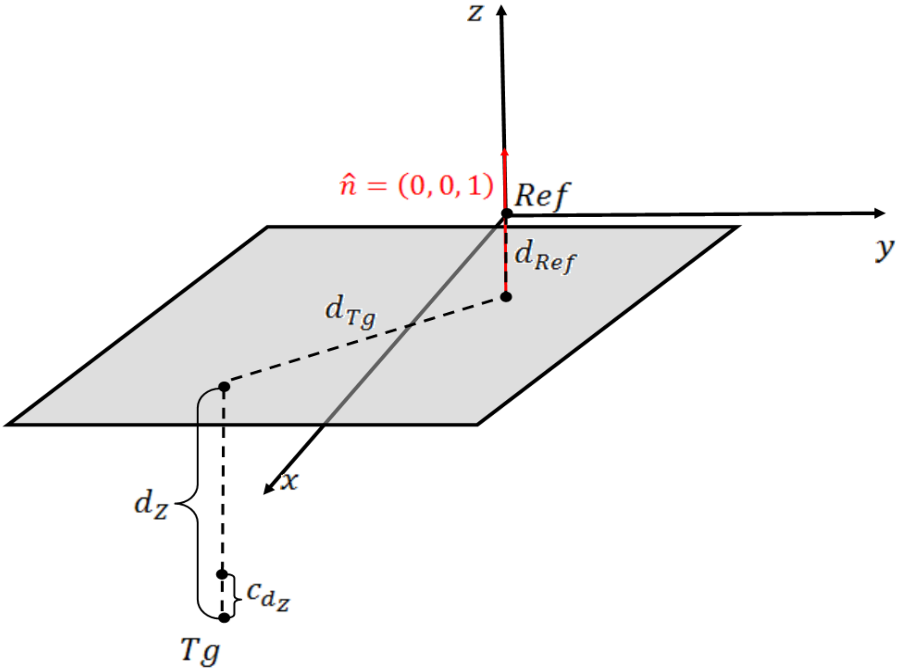
Simple objects (**not adjustable**):

- **Scalar** – for the unit vector $(i\ , j, k)$
- **Scalar –** for the standard deviations

4. VERTICAL DISTANCE MEASUREMENT MODELS

4.1 Vertical distance (DLEV) – Local Reference System

This measurement made in the Modified Local Astronomical System (MLA).



4.1.1 Mathematical Model

$$d_Z = 1 \cdot (Z_{Ref} - Z_{Tg}) + d_{ref} - c_{d_z} - d_{Tg}.\tan \varepsilon_{collimation}$$

Where

$\varepsilon_{collimation}$: Collimation angle

In this mathematical model we assume orientation of the normal vector $\hat{n}$ to be as is represented on a drawing above.

4.1.2 Partial Derivatives

For the Target coordinates

$$\frac{\partial d_z}{\partial *} = -\frac{\partial Z_{St}}{\partial *}$$

Where (2.9).

For the additional parameters:

$$\frac{\partial d_z}{\partial c_{d_z}} = -1$$

$$\frac{\partial d_z}{\partial d_{ref}} = 1$$

$$\frac{\partial d_z}{\partial \varepsilon_{collimation}} = -d.\,(1 + tan^2(\varepsilon_{collimation}))$$

$$\frac{\partial d_z}{\partial d_{Tg}} = -\tan \varepsilon_{collimation}$$

4.1.3 Stochastic model

$$\sigma_{dz}{}^2 = (\sigma_{vert\ dist} + d_{Tg} * \sigma_{ppm})^2 + \sigma_{offset\ of\ the\ staff}{}^2$$

4.1.4 Objects

Adjustable needed:

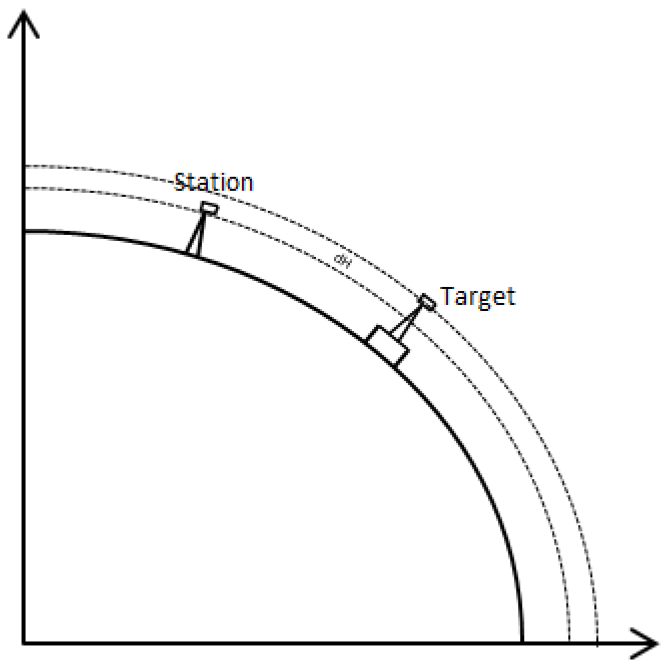
- **Plane** – consisting of:
  - **Point** – for the *Reference point*, fixed
  - **3Dvector** – known normal vector of the plane $\hat{n} = (0 \quad 0 \quad 1)$, fixed
  - **Scalar** – for the reference point distance (d_{Tg}) – variable
- **Point** – for the Station
- **Scalar** – for the constants c_{d_z} , $d_{ref}, \varepsilon_{collimation}$ and d_Z

Simple objects (**not adjustable**):

- **Scalar –** for the standard deviations

4.2 Vertical distance (DVER) – Geodetic Reference System

This measurement is made in the geodetic reference system, CGRF.



4.2.1 Mathematical Model

$$\mathrm{dH} = H_{Tg} - H_{St} - c_{dH}$$

4.2.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial \mathrm{dH}}{\partial *} = -\frac{\partial H_{St}}{\partial *} = \frac{\partial h_{St}}{\partial *}$$

Where (2.9).

And,

$$H_{St} = h_{St} - N$$

Where h_{St} is a third geodetic ellipsoidal coordinate (h) of the *station*. Cartesian coordinates of the point can be expressed as:

$$\begin{pmatrix} X^h \\ Y^h \\ Z^h \\ w \end{pmatrix} = \boldsymbol{A}^* \cdot \boldsymbol{M} \cdot \boldsymbol{x}_{0,St}^{\boldsymbol{h}}$$

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \frac{1}{w} \cdot \begin{pmatrix} X^h \\ Y^h \\ Z^h \end{pmatrix}$$

h_{St} Is then obtained from this Cartesian coordinates using equation (2.8).

For the Target coordinates:

$$\frac{\partial \mathrm{dH}}{\partial *} = \frac{\partial H_{Tg}}{\partial *} = \frac{\partial h_{Tg}}{\partial *}$$

Where (2.10).

And,

$$H_{Tg} = h_{Tg} - N$$

Where h_{St} is a third coordinate (h) of the *target* in the ellipsoidal coordinates. Cartesian coordinates of the point can be expressed as:

$$\begin{pmatrix} X^h \\ Y^h \\ Z^h \\ w \end{pmatrix} = \boldsymbol{A}^* \cdot \boldsymbol{M} \cdot \boldsymbol{x}_{0,Tg}^{\boldsymbol{h}}$$

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \frac{1}{w} \cdot \begin{pmatrix} X^h \\ Y^h \\ Z^h \end{pmatrix}$$

h_{Tg} Is then obtained from this Cartesian coordinates using equation (2.8).

For the additional parameters:

$$\frac{\partial \mathrm{dH}}{\partial c_{dH}} = -1$$

4.2.3 Stochastic model

$$\sigma_{dH}{}^{2} = \sigma_{m}{}^{2}$$

And σ_m represents the standard deviation of the observed value.

4.2.4 Objects

Adjustable needed:

- **Point** (with x and y coordinates excluded from the adjustment) – for *Station* and *Target*
- **Scalar** – for the c_{dH}

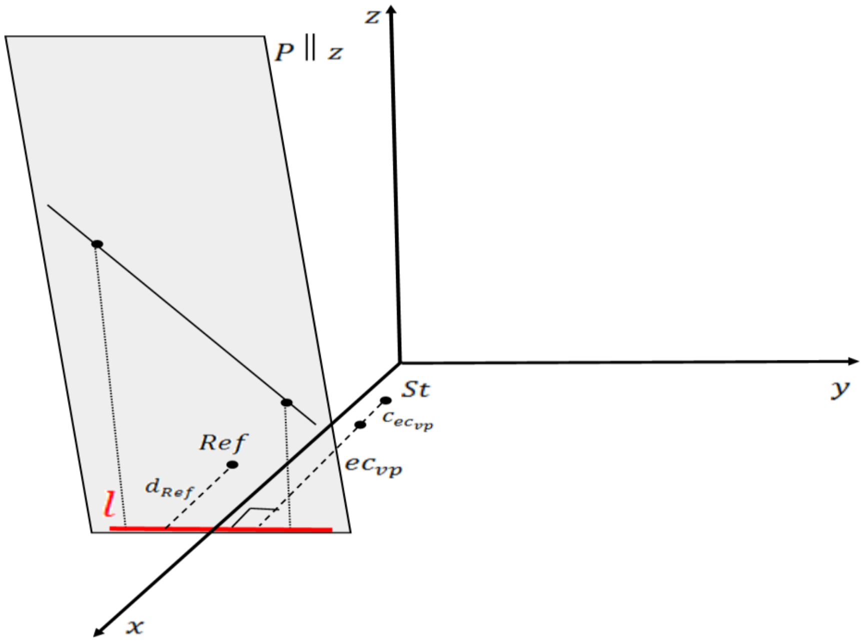
Simple objects (**not adjustable**):

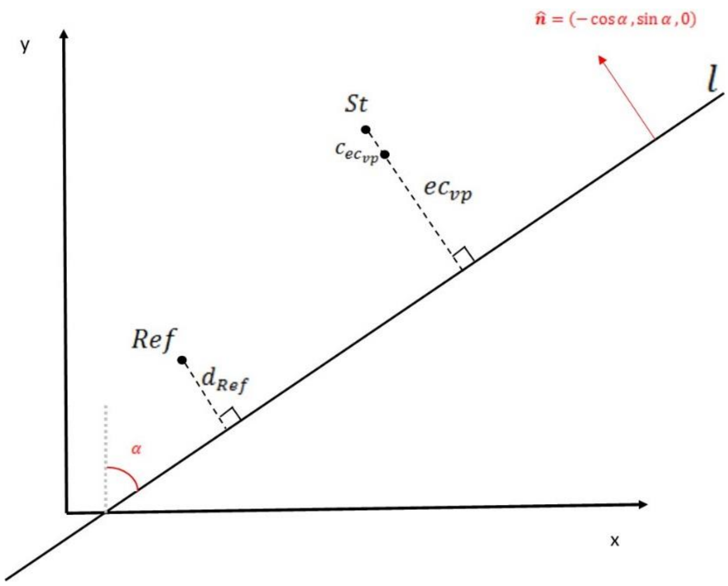
- **Scalar** – for the dH
- **Scalar –** for the standard deviations

5. OFFSET MEASUREMENT MODELS

5.1 Offset to a Vertical Plane (ECHO)

This measurement made in the Modified Local Astronomical System (MLA).





5.1.1 Mathematical Model

$$ec_{vp} = -\cos\alpha \cdot \left(X_{St} - X_{Ref}\right) + \sin\alpha \cdot \left(Y_{St} - Y_{Ref}\right) + d_{Ref} - c_{ec_{vp}}$$

In this mathematical model we assume orientation of the normal vector $\hat{n}$ to be as is represented on a drawing above.

5.1.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial ec_{vp}}{\partial *} = -\cos\alpha \cdot \frac{\partial X_{St}}{\partial *} + \sin\alpha \cdot \frac{\partial Y_{St}}{\partial *}$$

For the Reference coordinates:

$$\frac{\partial ec_{vp}}{\partial *} = \cos\alpha \cdot \frac{\partial X_{Ref}}{\partial *} - \sin\alpha \cdot \frac{\partial Y_{Ref}}{\partial *}$$

Where (2.9).

For the $\hat{n}$ vector elements:

$$\frac{\partial ec_{vp}}{\partial \alpha} = \sin\alpha\ \ (X_{St} - X_{Ref}) + \cos\alpha\ \ (Y_{St} - Y_{Ref})$$

For the Reference point distance:

$$\frac{\partial ec_{vp}}{\partial d_{Ref}} = 1$$

For the additional parameters:

$$\frac{\partial ec_{vp}}{\partial c_{ec_{vp}}} = -1$$

5.1.3 Stochastic model

$${\sigma_{ec_{vp}}}^2 = {\sigma_m}^2 + {\sigma_{CSt}}^2$$

And σ_m represents the standard deviation of the observed value and the ppm.

$$\sigma_m \quad = \sigma_{inst} \quad + ec * \sigma_{ppm}$$

5.1.4 Objects

Adjustable needed:

- **Plane** consisting of:
  - **Point** – for the *Reference point*
  - **3Dvector** – normal vector of the plane $\hat{n}$
  - **Scalar**– for the reference point distance d_{Ref} and the angle α

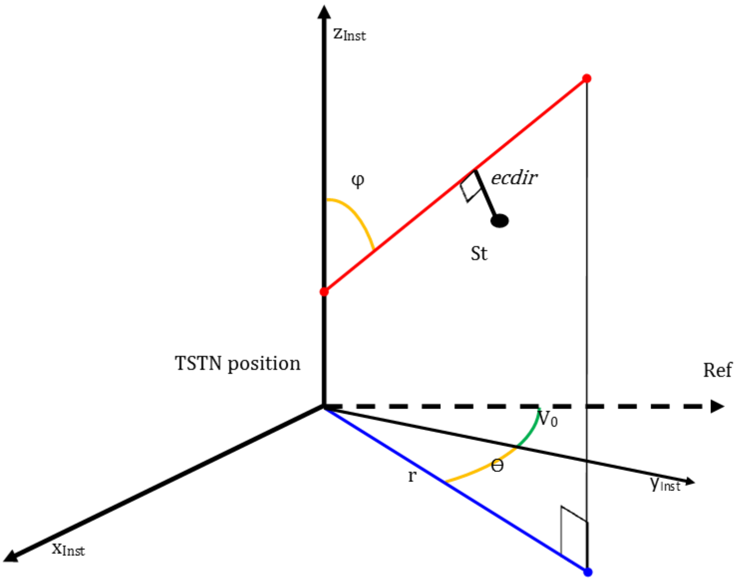
- **Point** (z-coordinate fixed, i.e. excluded from the adjustment) -- for the *Station*
- **Scalar**– for the $c_{ec_{vp}}$

Simple objects (**not adjustable**):

- **Scalar** – for ec_{vp}
- **Scalar**– for the standard deviations

5.2 Offset to a Spatial Line (ECDIR)

This measurement is made in the CERN Cartesian System (CCS). The line is define by a TSTN in L, the direction is given by TSTN measurement (θ is an observe horizontal angle and φ is an observed vertical angle).



5.2.1 Mathematical Model

$$ecdir = \left|(\mathbf{x_{St}} - \mathbf{x_{TSTN}}) \times \hat{\mathbf{l}}\right| - c_{ecdir}$$

Where

$$\mathbf{x_{St}} = \begin{pmatrix} X_{St} \\ Y_{St} \\ Z_{St} \end{pmatrix}, \mathbf{x_{TSTN}} = \begin{pmatrix} X_L \\ Y_L \\ Z_L \end{pmatrix}, \hat{\mathbf{l}} = \begin{pmatrix} -\sin(\theta + V0)\ \sin\varphi \\ \cos(\theta + V0)\ \sin\varphi \\ \cos\varphi \end{pmatrix} = \begin{pmatrix} l1 \\ l2 \\ l3 \end{pmatrix}$$

5.2.2 Partial Derivatives

$$\left\|(\mathbf{x_{St}} - \mathbf{x_{TSTN}}) \times \hat{\mathbf{l}}\right\|^2 = \|(\mathbf{x_{St}} - \mathbf{x_{TSTN}})\|^2 . \left\|\hat{\mathbf{l}}\right\|^2 - ((\mathbf{x_{St}} - \mathbf{x_{TSTN}}).\hat{\mathbf{l}})^2$$

We see that,

$$ecdir = \sqrt{\|(\mathbf{x_{St}} - \mathbf{x_{TSTN}})\|^2 - ((\mathbf{x_{St}} - \mathbf{x_{TSTN}}).\hat{\mathbf{l}})^2} - c_{ecdir}$$

Where

$$(\mathbf{x_{St}} - \mathbf{x_{TSTN}}).\hat{\mathbf{l}} = (X_{St} - X_{TSTN}) * l1 + (Y_{St} - Y_{TSTN}) * l2 + (Z_{St} - Z_{TSTN}) * l3 = pscal$$

For the Station coordinates:

$$\frac{\partial ecdir}{\partial *} = \frac{(X_{St} - X_{TSTN}) - l1 * \mathrm{pscal}}{ecdir} \cdot \frac{\partial X_{St}}{\partial *}$$

$$+\frac{(Y_{St}-Y_{TSTN})-l2*\text{pscal}}{ecdir}\cdot\frac{\partial Y_{St}}{\partial *}$$
$$+\frac{(Z_{St}-Z_{TSTN})-l3*\text{pscal}}{ecdir}\cdot\frac{\partial Z_{St}}{\partial *}$$

Where (2.9).

For the TSTN position:

$$\frac{\partial ecdir}{\partial *}=-\frac{\partial ecdir}{\partial *}$$

Where * can stand for: $x_{0,L};\ y_{0,L};\ z_{0,L};\ \omega_{j,L};\varphi_{j,L};\kappa_{j,L;}\, t_{j,1,L};t_{j,2,L};t_{j,3,L};l_{j,L}$.

For the additional parameters:

$$\frac{\partial ecdir}{\partial c_{ecdir}}=-1$$
$$\frac{\partial ecdir}{\partial V0}=-1*\frac{l2.(X_{St}-X_{TSTN})-l1.(Y_{St}-Y_{TSTN})}{ecdir}$$

5.2.3 Stochastic model

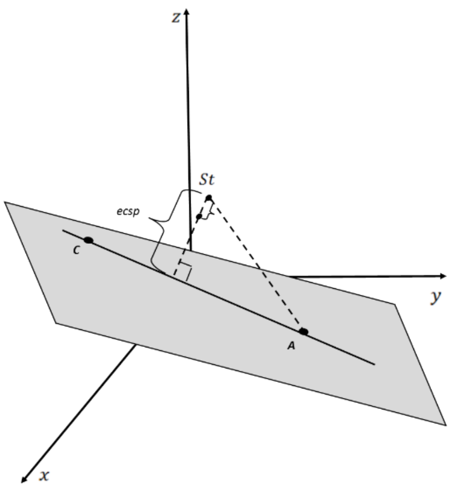
$$\sigma_{ecdir}{}^2=\sigma_m{}^2$$

And σ_m represents the standard deviation of the observed value and the ppm error

$$\sigma_m\ =\ \sigma_{obs}\ +ec\ *\sigma_{ppm}$$

5.3 Offset to a Spatial Line (ECSP)

This measurement is made in the CERN Cartesian System (CCS). The line is define by 2 points A and C. The instrument is stationned in B.



$$d=\left|\overrightarrow{AB}\right|$$
$$D=\left|\overrightarrow{AC}\right|$$
$$p=\overrightarrow{AB}.\overrightarrow{AC}$$
$$div=\left|\overrightarrow{AB}\wedge\overrightarrow{AC}\right|=\sqrt{D^2.d^2-p^2}$$

5.3.1 Mathematical Model

$$ec_{sl}=\sqrt{d^2-\left(\frac{p}{D}\right)^2}=\frac{1}{D}*div$$

5.3.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial ec_{sl}}{\partial *} = \frac{1}{div}\left(D * x_{AB} - \frac{p}{D} * x_{AC}\right) \cdot \frac{\partial X_{St}}{\partial *} + \frac{1}{div}\left(D * y_{AB} - \frac{p}{D} * y_{AC}\right) \cdot \frac{\partial Y_{St}}{\partial *} + \frac{1}{div}\left(D * z_{AB} - \frac{p}{D} * z_{AC}\right) \cdot \frac{\partial Z_{St}}{\partial *}$$

For the first point on the line:

$$\frac{\partial ec_{sl}}{\partial *} = \frac{1}{div}\left(-D * x_{AB} - \frac{p}{D} * (x_{AC} + x_{AB}) - \frac{p^2}{D^3} * x_{AC}\right) \cdot \frac{\partial X_A}{\partial *} + \frac{1}{div}\left(-D * y_{AB} - \frac{p}{D} * (y_{AC} + y_{AB}) - \frac{p^2}{D^3} * y_{AC}\right) \cdot \frac{\partial Y_A}{\partial *} + \frac{1}{div}\left(-D * z_{AB} - \frac{p}{D} * (z_{AC} + z_{AB}) - \frac{p^2}{D^3} * z_{AC}\right) \cdot \frac{\partial Z_A}{\partial *}$$

For the second point on the line:

$$\frac{\partial ec_{sl}}{\partial *} = \frac{1}{div}\left(-\frac{p}{D} * x_{AB} + \frac{p^2}{D^3} * x_{AC}\right) \cdot \frac{\partial X_C}{\partial *} + \frac{1}{div}\left(-\frac{p}{D} * y_{AB} + \frac{p^2}{D^3} * y_{AC}\right) \cdot \frac{\partial Y_C}{\partial *} + \frac{1}{div}\left(-\frac{p}{D} * z_{AB} + \frac{p^2}{D^3} * z_{AC}\right) \cdot \frac{\partial Z_C}{\partial *}$$

Where * can stand for: $x_{0,L};\ y_{0,L};\ z_{0,L};\ \omega_{j,L}; \varphi_{j,L}; \kappa_{j,L;}\ t_{j,1,L}; t_{j,2,L}; t_{j,3,L}; l_{j,L}$.

5.3.3 Stochastic model

$$\sigma_{ec_{sl}}{}^2 = \sigma_m{}^2$$

And σ_m represents the standard deviation of the observed value and the ppm error

$$\sigma_m \;=\; \sigma_{obs} \;+ ec \;* \sigma_{ppm}$$

5.3.4 Objects

Adjustable needed:

- **Points** – for the *Station, and the line definition*
- **Scalar** – for the $c_{ec_{sl}}$

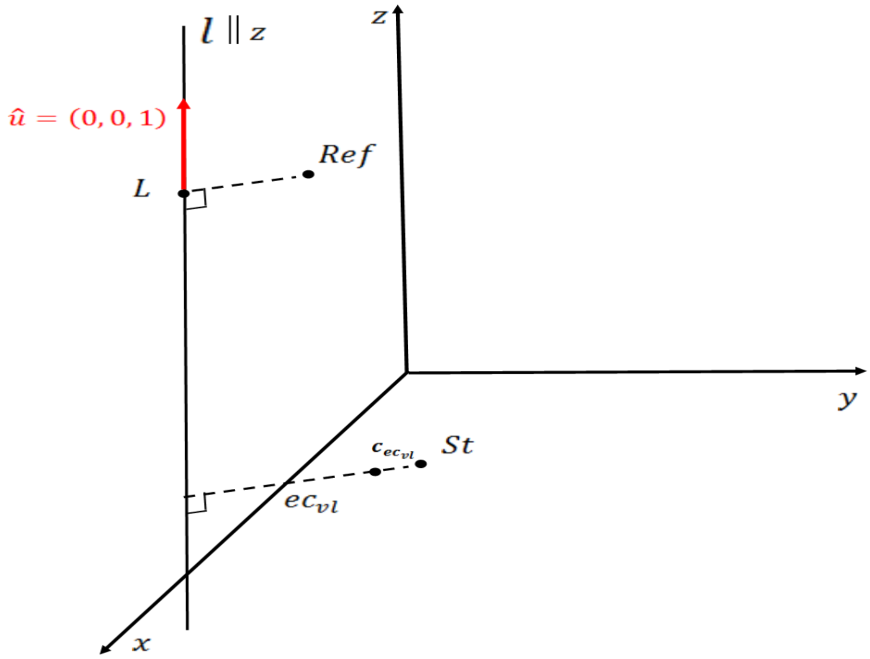
Simple objects (**not adjustable**):

- **Scalar** – for the ec_{sl}
- **Scalar –** for the standard deviations

5.4 Offset to a Vertical Line (ECVE)

This measurement is made in the Modified Local Astronomical System (MLA).

The point on the line can be known / measured or unknown. In the first case the reference point is the same as L. In the second case, L is the mean of all estimated stationed point (AdjustablePoint in X and Y) and Ref is the mean off all provisional St points. Fix the Z coordinate allow to not add a constraint for the measurement.



5.4.1 Mathematical Model

$$ec_{vl} = \sqrt{(X_L - X_{St})^2 + (Y_L - Y_{St})^2} - c_{ec_{vl}}$$

5.4.2 Partial Derivatives

Let

$$D = \sqrt{(X_L - X_{St})^2 + (Y_L - Y_{St})^2}$$

For the Station coordinates:

$$\frac{\partial ec_{vl}}{\partial *} = \frac{X_{St} - X_L}{D} \cdot \frac{\partial X_{St}}{\partial *} + \frac{Y_{St} - Y_L}{D} \cdot \frac{\partial Y_{St}}{\partial *}$$

For the line point coordinates:

$$\frac{\partial ec_{vl}}{\partial *} = \frac{X_L - X_{St}}{D} \cdot \frac{\partial X_L}{\partial *} + \frac{Y_L - Y_{St}}{D} \cdot \frac{\partial Y_L}{\partial *}$$

Where * can stand for: $x_{0,L}$; $y_{0,L}$; $z_{0,L}$; $\omega_{j,L}$; $\varphi_{j,L}$; $\kappa_{j,L;}$ $t_{j,1,L}$; $t_{j,2,L}$; $t_{j,3,L}$; $l_{j,L}$.

For the additional parameters:

$$\frac{\partial ec_{vl}}{\partial c_{ec_{vl}}} = -1$$

5.4.3 Stochastic model

$$\sigma_{ec_{vl}}{}^2 = \sigma_m{}^2 + \frac{(Y_{St} - Y_L)^2 + (X_{St} - X_L)^2}{D^2} \sigma_{CSt}{}^2$$

And σ_m represents the standard deviation of the observed value and the ppm error

$$\sigma_m = \sigma_{obs} + ec * \sigma_{ppm}$$

5.4.4 Objects

Adjustable:

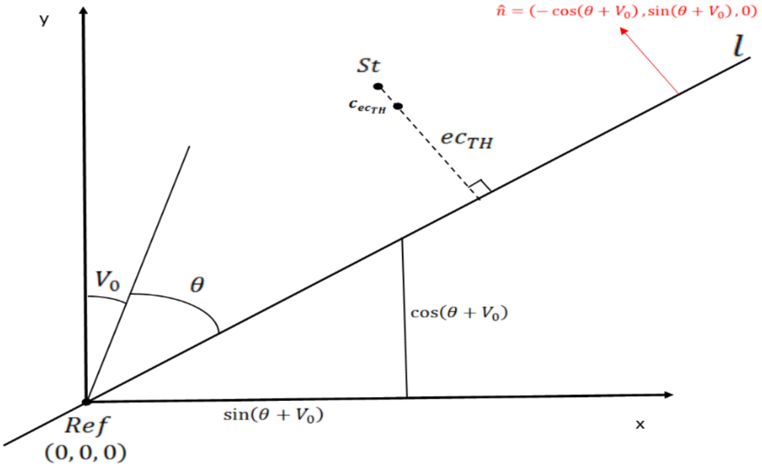
- **Point** – for the *Station*
- **Line**, consisting of:
  - **Point** – line point **L**
  - **3D vector** – line vector **û** is known (0,0,1)
  - **Point** – *reference point*
- **Scalar** – for a $c_{ec_{vl}}$

Simple objects:

- **Scalar**– for a ec_{vl}
- **Scalar**– for the standard deviations

5.5 Offset to a Theodolite Plane (ECTH)

This measurement is made in the Modified Local Astronomical System (MLA).



5.5.1 Mathematical Model

$$ec_{TH} = -\cos(\theta + V_0) \cdot (X_{St} - X_{Ref}) + \sin(\theta + V_0) \cdot (Y_{St} - Y_{Ref}) - c_{ec_{TH}}$$

5.5.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial ec_{TH}}{\partial *} = -\cos(\theta + V_0) \cdot \frac{\partial X_{St}}{\partial *} + \sin(\theta + V_0) \cdot \frac{\partial Y_{St}}{\partial *}$$

Where (2.9).

For the Reference point coordinates:

$$\frac{\partial ec_{TH}}{\partial *} = \cos(\theta + V_0) \cdot \frac{\partial X_{Ref}}{\partial *} - \sin(\theta + V_0) \cdot \frac{\partial Y_{Ref}}{\partial *}$$

Where * can stand for: $x_{0,Ref}$; $y_{0,Ref}$; $z_{0,Ref}$; $\omega_{j,Ref}$; $\varphi_{j,Ref}$; $\kappa_{j,Ref;}$ $t_{j,1,Ref}$; $t_{j,2,Ref}$; $t_{j,3,Ref}$; $l_{j,Ref}$.

For the angles:

$$\frac{\partial ec_{TH}}{\partial V_0} = \sin(\theta + V_0) \cdot \left(X_{St} - X_{Ref}\right) + \cos(\theta + V_0) \cdot \left(Y_{St} - Y_{Ref}\right)$$

For the additional parameters:

$$\frac{\partial ec_{TH}}{\partial c_{ec_{TH}}} = -1$$

5.5.3 Stochastic model

$$\sigma_{ec_{TH}}{}^2 = \sigma_m{}^2 + (\cos(\theta + V_0)^2 + \sin(\theta + V_0)^2)\, \sigma_{CSt}{}^2$$

And σ_m represents the standard deviation of the measured offset and the ppm error

$$\sigma_m \;=\; \sigma_{meas\_offset} \;+ ec \;* \sigma_{ppm}$$

5.5.4 Objects

Adjustable:

- **Point** – for the *Station* and *Reference point*
- **Angle** – for the V_0
- **Scalar** – for the $c_{ec_{TH}}$

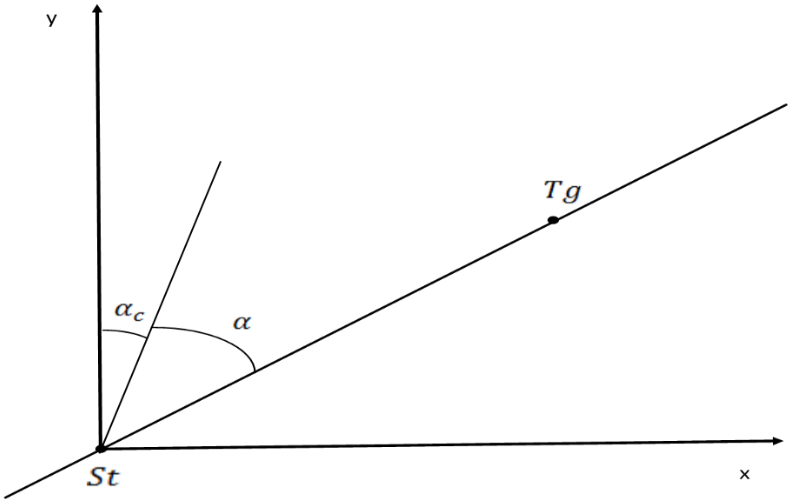
Simple objects:

- **Angle** – for the θ
- **Scalar** – for the ec_{TH}
- **Scalar** – for the standard deviations

6. ADDITIONAL MEASUREMENT MODELS

6.1 5.1 Gyro-Theodolite Azimuth (ORIE)

This measurement is made in the Local Astronomical System (LA) and, therefore, it involves different transformation than most of the other observation equations, which are defined in MLA.



6.1.1 Mathematical Model

$$\alpha = \tan^{-1}\left(\frac{X_{Tg} - X_{St}}{Y_{Tg} - Y_{St}}\right) - \alpha_c$$

6.1.2 Partial Derivatives

For the Station coordinates:

$$\frac{\partial \alpha}{\partial *} = -\frac{\left(Y_{Tg} - Y_{St}\right)}{D^2} \cdot \frac{\partial X_{St}}{\partial *} + \frac{\left(X_{Tg} - X_{St}\right)}{D^2} \cdot \frac{\partial Y_{St}}{\partial *}$$

Where

$$D^2 = \left(X_{Tg} - X_{St}\right)^2 + \left(Y_{Tg} - Y_{St}\right)^2$$

Where (2.9).

For the Target coordinates:

$$\frac{\partial \alpha}{\partial *} = \frac{\left(Y_{Tg} - Y_{St}\right)}{D^2} \cdot \frac{\partial X_{Tg}}{\partial *} + \frac{\left(X_{St} - X_{Tg}\right)}{D^2} \cdot \frac{\partial Y_{Tg}}{\partial *}$$

Where (2.10).

For the additional parameters:

$$\frac{\partial \alpha}{\partial \alpha_c} = -1$$

6.1.3 Stochastic model

$$\sigma_\alpha{}^2 = \sigma_m{}^2 + \sigma_{\alpha c}{}^2 + \frac{\left(Y_{Tg} - Y_{St}\right)^2 + \left(X_{St} - X_{Tg}\right)^2}{D^4}\left(\sigma_{CTg}{}^2 + \sigma_{CSt}{}^2\right)$$

By default $\sigma_{\alpha c}{}^2 = 0$
And σ_m is the standard deviation of the observed value

6.1.4 Objects

Adjustable:

- **Point** – for the *Station* and *Target*
- **Angle** – for α_c

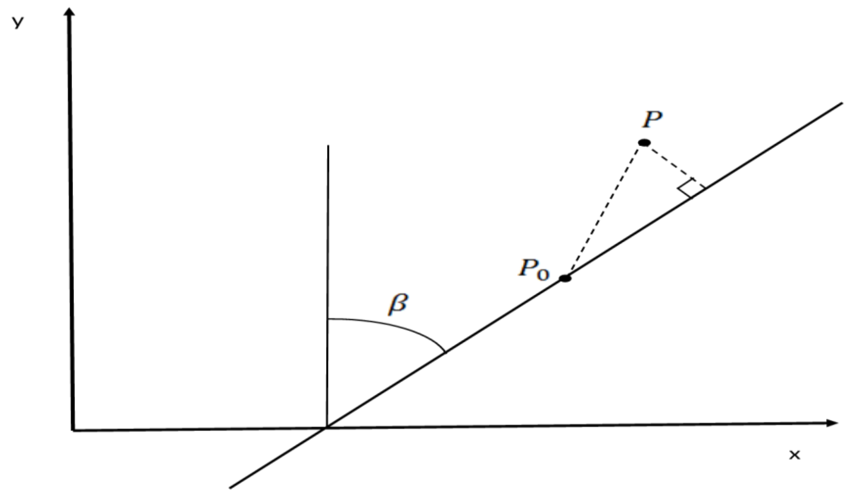
Simple objects:

- **Angle** – for α
- **Scalar –** for the standard deviations

7. CONSTRAINT MODELS

7.1 Radial Offset Constraint (RADI)

This measurement is made in the CERN Cartesian System (CCS). P0 is the provisional point and P is the estimated point.



7.1.1 Mathematical Model

$$F = -\cos(\beta) \cdot \left(X_P - X_{P_0}\right) + \sin(\beta) \cdot \left(Y_P - Y_{P_0}\right) = 0$$

7.1.2 Partial Derivatives

For the point P coordinates:

$$\frac{\partial F}{\partial *} = -\cos(\beta) \cdot \frac{\partial X_P}{\partial *} + \sin(\beta) \cdot \frac{\partial Y_P}{\partial *}$$

Where * can stand for: $x_{0,P}$; $y_{0,P}$; $z_{0,P}$; $\omega_{j,P}$; $\varphi_{j,P}$; $\kappa_{j,P;}$ $t_{j,1,P}$; $t_{j,2,P}$; $t_{j,3,P}$; $l_{j,P}$.

7.1.3 Stochastic model

$$\sigma_F{}^2 = \sigma_{CP}{}^2$$

7.1.4 Objects

Adjustable:

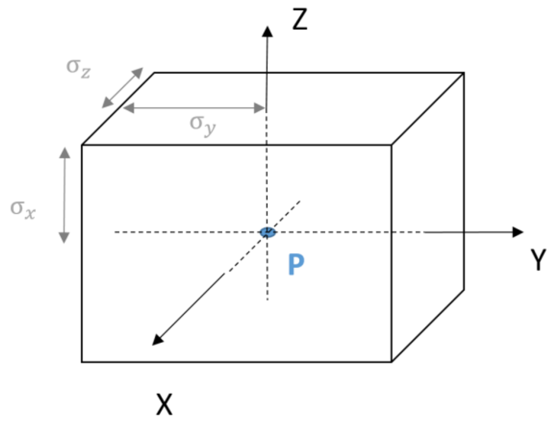
- **Point** – $\boldsymbol{P_0}$ and $\boldsymbol{P}$ and points

Simple objects:

- **Scalar –** for the standard deviations
- **Angle** – β angle

7.2 Observed coordinates (OBSXYZ)

This measurement is made in a 3D Cartesian system. The points is known / constraint in 3D space. Point coordinates can move in a box, sized by its precisions (σx, σy, σz).



7.2.1 Mathematical Model

$$X_P - X_{obs} = 0$$
$$Y_P - Y_{obs} = 0$$
$$Z_P - Z_{obs} = 0$$

7.2.2 Partial Derivatives

For the point P coordinates:

$$\frac{\partial F}{\partial *} = 1$$

Where * can stand for: $x_{0,P};\ y_{0,P};\ z_{0,P};\ \omega_{j,P}; \varphi_{j,P}; \kappa_{j,P;}\ t_{j,1,P}; t_{j,2,P}; t_{j,3,P}; l_{j,P}$.

7.2.3 Stochastic model

$$\sigma_F{}^2 = \sigma_{CP}{}^2$$

7.2.4 Objects

Adjustable:
- **Point –** ***P***

Simple objects:
- **Scalar –** for the standard deviations

7.3 Orientation Constraint (PDOR)

This measurement is made in the CERN Cartesian System (CCS).

7.3.1 Mathematical Model

$$\beta = \tan^{-1}\left(\frac{X_P - X_{CALA}}{Y_P - Y_{CALA}}\right)$$

7.3.2 Partial Derivatives

For the point P coordinates:

$$\frac{\partial \beta}{\partial *} = \frac{(Y_P - Y_{CALA})}{D^2} \cdot \frac{\partial X_P}{\partial *} + \frac{(X_{CALA} - X_P)}{D^2} \cdot \frac{\partial Y_P}{\partial *}$$

Where

$$D^2 = (X_P - X_{CALA})^2 + (Y_P - Y_{CALA})^2$$

7.3.3 Objects

Adjustable needed:

- **Point** – for the P point
- **Point** – for the $\boldsymbol{x_{CALA}}$ point (all coordinates fixed)

Simple objects (**not adjustable**):

- **Angle** – for the β
- **Scalar –** for the standard deviations

Where * can stand for: $x_{0,P};\ y_{0,P};\ z_{0,P};\ \omega_{j,P}; \varphi_{j,P}; \kappa_{j,P;}\ t_{j,1,P}; t_{j,2,P}; t_{j,3,P}; l_{j,P}$.

7.4 Point Coordinate

This measurement is given in the CCS or one of the Local Object Reference (LOR) systems.

$$\boldsymbol{X}_P = \boldsymbol{X}_{P,0}$$

7.4.1 Mathematical Model

<u>In 3D space:</u>

$$F = (X_P - X_{P,0})^2 + (Y_P - Y_{P,0})^2 + (Z_P - Z_{P,0})^2 = 0$$

<u>In 2D space:</u>

$$F = (X_P - X_{P,0})^2 + (Y_P - Y_{P,0})^2 = 0$$

<u>In 1D space:</u>

$$F = (Z_P - Z_{P,0})^2 = 0$$

7.4.2 Partial Derivatives

In 3D space:

$$\frac{\partial F}{\partial *} = 2 \cdot (X_P - X_{P,0}) \cdot \frac{\partial X_P}{\partial *} + 2 \cdot (Y_P - Y_{P,0}) \cdot \frac{\partial Y_P}{\partial *} + 2 \cdot (Z_P - Z_{P,0}) \cdot \frac{\partial Z_P}{\partial *}$$

Where * can stand for: $x_{0,P};\ y_{0,P};\ z_{0,P};\ \omega_{j,P}; \varphi_{j,P}; \kappa_{j,P;}\, t_{j,1,P}; t_{j,2,P}; t_{j,3,P}; l_{j,P}$.

For the measured coordinates,

$$\frac{\partial F}{\partial X_{P,0}} = -2 \cdot (X_P - X_{P,0})$$

$$\frac{\partial F}{\partial Y_{P,0}} = -2 \cdot (Y_P - Y_{P,0})$$

$$\frac{\partial F}{\partial Z_{P,0}} = -2 \cdot (Z_P - Z_{P,0})$$

In 2D space:

$$\frac{\partial F}{\partial *} = 2 \cdot (X_P - X_{P,0}) \cdot \frac{\partial X_P}{\partial *} + 2 \cdot (Y_P - Y_{P,0}) \cdot \frac{\partial Y_P}{\partial *}$$

Where * can stand for: $x_{0,P};\ y_{0,P};\ \omega_{j,P}, t_{j,1,P}; t_{j,2,P}; l_{j,P}$.

For the measured coordinates,

$$\frac{\partial F}{\partial X_{P,0}} = -2 \cdot (X_P - X_{P,0})$$

$$\frac{\partial F}{\partial Y_{P,0}} = -2 \cdot (Y_P - Y_{P,0})$$

In 1D space:

$$\frac{\partial F}{\partial *} = 2 \cdot (Z_P - Z_{P,0}) \cdot \frac{\partial Z_P}{\partial *}$$

Where * can stand for: $z_{0,P}; t_{j,1,P}; l_{j,P}$.

For the measured coordinate,

$$\frac{\partial F}{\partial Z_{P,0}} = -2 \cdot (Z_P - Z_{P,0})$$

7.4.3 Stochastic model

For point coordinate, standard deviation are used to fill in the weight matrix of unknowns.

For X and Y components, the value is $\sigma = \frac{1}{2}\sigma_{CP}$
And 0 for the Z component.

7.4.4 Objects

- **(3D, 2D and 1D) points**

8. REFERENCES

[1] JONES, M. (1999). THE RXRYRZ MATRIX. GENEVA: EST-SU INTERNAL NOTE, EDMS DOCUMENT NO. 107903, CERN.

[2] JONES, M. (2000). SPATIAL TRANSFORMATIONS: TRANSFORMATIONS BETWEEN THE GEODETIC AND ASTRONOMICAL REFERENCE & COORDINATE SYSTEMS. GENEVA: EST-SU INTERNAL NOTE, EDMS DOCUMENT NO. 107907, CERN.

[3] SOLER, T., & CHIN, M. (1985). ON TRANSFORMATION OF COVARIANCE MATRICES BETWEEN LOCAL CARTESIAN SYSTEMS AND COMMUTATIVE DIAGRAMS. WASHINGTON, D.C.: ASP-ACSM.

[4] ZWIENER, J. (2011). 3-D INTEGRATED NETWORK ADJUSTMENT WITH A PARAMETRIC HEIGHT REFERENCE SURFACE.